

Statistics for Public Health: Descriptive Statistics, Probability and Inferential Statistics: Study Notes

Oct 6, 2026 · Md. Ashiqur Rahman Khan

How to use these notes

These notes cover the first three topics of the Public Health Data Analyst roadmap. Every idea follows the same order: a definition in simple words, the formula, a worked example with the steps shown, and what it means for a public health decision. There is no code in this document. Every calculation can be followed with a basic calculator.

Part	What it answers	After it you should be able to
1. Descriptive statistics	What does my data look like? How do I summarise thousands of records in a few honest numbers?	Pick the right summary for any variable, and build a correct "Table 1".
2. Probability and distributions	How likely is an event? How do I read a diagnostic test? Which distribution describes my data?	Calculate risk, apply Bayes' theorem to a screening test, and match data to a distribution.
3. Inferential statistics	What can a sample tell me about the whole population, and how sure can I be?	Build a confidence interval, choose and interpret a test, and plan a sample size.

How the examples are labelled

Label	Meaning
Real data	The numbers come from a published source, listed in "Sources" at the end. Examples: the 2023 dengue outbreak in Bangladesh, the Bangladesh Demographic and Health Survey 2022, the 1954 Salk polio vaccine trial.
Illustrative numbers	A realistic public health situation with made-up numbers chosen so the arithmetic is easy to follow. Do not quote these as facts.

The four coloured boxes

IN SIMPLE WORDS

The plain definition.
Read this first.

EXAMPLE

A worked calculation
or a real case.

COMMON MISTAKE

An error seen often in
theses and reports.

REMEMBER

The one line to keep.

Symbols you will see

Statistics uses Greek letters for the population (the truth we cannot see) and Latin letters for the sample (the data we have).

Quantity	Population (parameter)	Sample (statistic)	How to read it
Size	N	n	Number of people or units
Mean	μ (mu)	$\bar{x}$ (x-bar)	The average
Standard deviation	σ (sigma)	s	Typical distance from the mean
Variance	σ^2	s^2	Standard deviation squared
Proportion	p or π	$\hat{p}$ (p-hat)	Share of people with the characteristic
Correlation	ρ (rho)	r	Strength of a straight-line relationship
Sum	Σ (capital sigma)		"Add up all the values"
Error rates	α (alpha), β (beta)		Chance of a false alarm, chance of a missed effect

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Part 1: Descriptive Statistics

Descriptive statistics turns raw records into a small number of summaries that a person can read and act on. It makes no claim beyond the data in front of you. Every report, dashboard and research paper in public health starts here.

1.1 What descriptive statistics is and why it matters

IN SIMPLE WORDS

Descriptive statistics means describing the data you have: how many, how much, how spread out, and what shape. It answers "what happened?" It does not answer "why?" or "will it happen again?"

A surveillance officer who receives 321,179 dengue records cannot read them one by one. Descriptive statistics lets the officer say: how many cases, in which months, in which age groups, in which districts, and how many died. In epidemiology this is called describing disease by **person, place and time**.

Question	What you summarise	Public health example
Person	Age, sex, occupation, education, wealth	Which age group has the most dengue deaths?
Place	Division, district, urban or rural, ward	Is stunting higher in Sylhet than in Khulna?
Time	Day, week, month, year, season	In which month do cases peak?

1.2 Population, sample, parameter and statistic

IN SIMPLE WORDS

- **Population:** everyone you want to know about. Example: all children under 5 in Bangladesh.
- **Sample:** the part of the population you actually measured. Example: the children measured in a survey.
- **Parameter:** a true number about the population. It is usually unknown. Example: the true percentage of all under-5 children who are stunted.
- **Statistic:** a number calculated from the sample. Example: 24% stunted among the children measured.

REAL EXAMPLE Real data

The Bangladesh Demographic and Health Survey (BDHS) 2022 interviewed 30,018 households and 30,078 women aged 15 to 49. Nobody measured every child in the country. The published figure, 24% of under-5 children stunted, is a **statistic** from the sample that is used to estimate the **parameter** for the whole country. Part 3 explains how far a statistic can be trusted.

1.3 Types of variables

A **variable** is anything that differs from one person to the next: age, sex, blood pressure, vaccination status. The type of variable decides which summary, which chart and which statistical test you are allowed to use, so identify the type before anything else.

Type	Meaning in simple words	Public health examples	Correct summary
Nominal (categorical)	Named groups with no order	Sex, blood group, division, cause of death	Count and percentage, mode
Binary (a nominal variable with two groups)	Yes or no	Vaccinated or not, died or survived, test positive or negative	Count and percentage
Ordinal (categorical)	Groups with a natural order, but the gaps between groups are not equal	Education level, wealth quintile, pain score, cancer stage	Count and percentage, median
Discrete (numerical)	Whole-number counts	Number of children, number of clinic visits, number of cases per week	Mean or median, with a spread; or rate
Continuous (numerical)	Measured on a scale; any value is possible	Weight, height, haemoglobin, blood pressure, age	Mean and SD, or median and IQR

Two more labels are used in every analysis:

- **Outcome** (dependent variable): the result you care about, such as having diabetes.
- **Exposure** or **predictor** (independent variable): the thing that may influence the outcome, such as BMI or smoking.

COMMON MISTAKE

Codes are not numbers. If sex is stored as 1 = male and 2 = female, a "mean sex of 1.48" has no meaning. The same applies to division codes and to ordinal scales such as education level (1 to 5). A second mistake is cutting a continuous variable into groups too early, for example turning age into "young" and "old". Grouping throws away information. Keep the original values and group only for presentation.

1.4 Summarising categorical data: counts, proportions, ratios and rates

For a categorical variable the basic summary is a **frequency table**: each category, how many people fall in it, and what percentage that is. Public health then builds four kinds of number from counts. They are often confused, even in published reports.

Measure	Meaning in simple words	Formula	Example
Proportion	A part divided by the whole. The top number is included in the bottom number. Always between 0 and 1.	$\frac{a}{a+b}$	Stunted children ÷ all children measured = 0.24
Percentage	A proportion multiplied by 100.	$\frac{a}{a+b} \times 100$	24% of under-5 children stunted (BDHS 2022)
Ratio	One number divided by a different number. The top is not part of the bottom.	$\frac{a}{b}$	Stunted to not stunted = 24 : 76, about 1 to 3.2
Rate	How fast events happen: events divided by the population at risk over a period of time.	$\frac{\text{events}}{\text{population} \times \text{time}} \times k$	Dengue cases per 100,000 people per year

The multiplier k (100, 1,000 or 100,000) only makes small numbers readable. Use 100 for common events, 1,000 for births and deaths, and 100,000 for rarer diseases.

REAL EXAMPLE: DENGUE IN BANGLADESH, 2023 Real data

The Directorate General of Health Services (DGHS) recorded 321,179 hospitalised dengue cases and 1,705 deaths in 2023, the largest outbreak in the country's history.

- **Proportion (case fatality):** $1,705 \div 321,179 = 0.0053$, or **0.53%**. Out of every 1,000 hospitalised patients, about 5 died. This is called the case fatality "rate" by habit, but it is a proportion, because the deaths are part of the cases.
- **Rate (incidence):** with a population of about 170 million (2022 census), $321,179 \div 170,000,000 \times 100,000 \approx$ **189 hospitalised cases per 100,000 people in the year**.
- **Percentage (seasonality):** 303,913 of the cases were recorded from July to November, which is **94.6%** of the year's total in five months.

These three numbers already tell a health manager what to prepare (beds for a July to November surge) and how deadly the disease was among those admitted.

COMMON MISTAKE

Comparing raw counts between places of different size. Dhaka will almost always have more cases than a small district simply because more people live there. Compare **rates per population**, not counts. Also state the denominator clearly: the 0.53% above is deaths among *hospitalised* cases. Mild cases that never reached a hospital are not counted, so the true fatality among all infections is lower.

Some named public health indicators that follow these definitions:

Indicator	How it is calculated	Real value
Under-5 mortality rate	Deaths before age 5 per 1,000 live births	31 per 1,000 (BDHS 2022)
Prevalence of stunting	Stunted children ÷ children measured × 100	24% (BDHS 2022)
Caesarean section rate	C-section births ÷ all births × 100	45% (BDHS 2022)
Case fatality	Deaths from a disease ÷ cases of that disease × 100	0.53% (dengue 2023, hospitalised)

1.5 Measures of centre: mean, median and mode

A measure of centre gives one "typical" value for a numerical variable.

IN SIMPLE WORDS

- **Mean:** add all the values and divide by how many there are. It is the balance point of the data.
- **Median:** put the values in order and take the middle one. Half the values are below it and half are above. With an even number of values, average the two middle ones.
- **Mode:** the value that appears most often. It is the only measure of centre that works for nominal data.

$$\bar{x} = \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

WORKED EXAMPLE: LENGTH OF HOSPITAL STAY

Illustrative numbers

Ten dengue patients stayed in hospital for these numbers of days: 2, 2, 3, 3, 3, 4, 4, 5, 6, 28.

- **Mean** = $(2 + 2 + 3 + 3 + 3 + 4 + 4 + 5 + 6 + 28) \div 10 = 60 \div 10 = \mathbf{6.0 \text{ days}}$
- **Median** = average of the 5th and 6th values = $(3 + 4) \div 2 = \mathbf{3.5 \text{ days}}$
- **Mode** = **3 days** (appears three times)

Nine of the ten patients stayed 6 days or fewer, yet the mean is 6.0. One patient with a 28-day stay pulled the mean upward. The median, 3.5 days, describes the typical patient better. A hospital planning beds needs both: the median for the typical patient and the mean for total bed-days (mean $\times$ number of patients).

Which one should you report?

Situation	Report	Why
Numerical data, roughly symmetric, no extreme values	Mean (with SD)	Uses every value and works with most statistical tests
Numerical data that is skewed or has outliers: income, length of stay, cost, incubation period, viral load	Median (with IQR)	Not pulled by extreme values
Ordinal data: pain score, satisfaction, wealth quintile	Median	The gaps between categories are not equal, so a mean is not meaningful
Nominal data: blood group, cause of death	Mode, or the full percentage table	Categories cannot be ordered or added

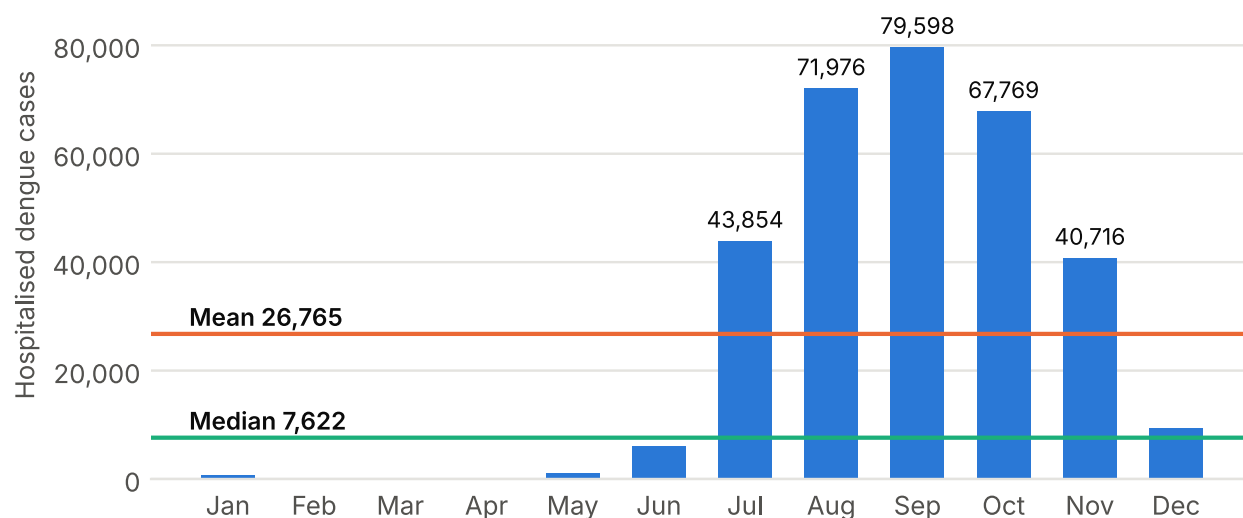


Figure 1.1. Monthly hospitalised dengue cases, Bangladesh, 2023 (real data, DGHS). The mean of the 12 months is 26,765 cases, but the median is only 7,622. Seven months lie below the mean. For data shaped like this, the mean describes almost no actual month. This bar chart of cases over time is called an epidemic curve.

Two special means used in public health

Weighted mean. When groups have different sizes, each group's value must count in proportion to its size.

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

WORKED EXAMPLE: VACCINATION COVERAGE

Illustrative numbers

Three upazilas report measles vaccination coverage of 90%, 80% and 60%. Their child populations are 10,000, 30,000 and 60,000.

- Simple mean = $(90 + 80 + 60) \div 3 = 76.7\%$. This is wrong for the district as a whole.
- Weighted mean = $(0.90 \times 10,000 + 0.80 \times 30,000 + 0.60 \times 60,000) \div 100,000 = 69,000 \div 100,000 = 69\%$.

The largest upazila has the lowest coverage, so the true district coverage is well below the simple average. National survey estimates such as BDHS are weighted means for the same reason.

Geometric mean. Used when values multiply instead of add, or when they span several powers of ten: antibody titres, bacterial counts, viral load. Take the logarithm of each value, find the ordinary mean of the logs, then convert back.

$$GM = \sqrt[n]{x_1 \times x_2 \times \cdots \times x_n} = 10^{\text{mean of } \log_{10}(x_i)}$$

WORKED EXAMPLE: ANTIBODY TITRES

Illustrative numbers

Five vaccinated people have antibody titres of 10, 20, 40, 80 and 640. The logs (base 10) are 1.000, 1.301, 1.602, 1.903 and 2.806. Their mean is 1.722, and $10^{1.722} = 52.8$. The ordinary mean is 158, which is higher than four of the five values. The geometric mean titre (GMT) of 52.8 is the standard summary in vaccine studies.

1.6 Measures of spread: range, variance, standard deviation and IQR

Two districts can have the same mean blood pressure while one has everybody near the mean and the other has many very low and very high values. A measure of centre alone hides this. Always report a measure of spread with it.

IN SIMPLE WORDS

- **Range:** largest value minus smallest value.
- **Variance:** the average of the squared distances from the mean.
- **Standard deviation (SD):** the square root of the variance. It is the typical distance of a value from the mean, in the same unit as the data.
- **Interquartile range (IQR):** the width of the middle 50% of the data, from the 25th percentile (Q1) to the 75th percentile (Q3).

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$s = \sqrt{s^2}$$

$$\text{IQR} = Q_3 - Q_1$$

WORKED EXAMPLE: SYSTOLIC BLOOD PRESSURE*Illustrative numbers*

Eight adults at a community clinic have systolic blood pressure (mmHg) of 110, 118, 122, 126, 130, 134, 140 and 160. The mean is $1,040 \div 8 = 130$.

Value x	110	118	122	126	130	134	140	160	Total
Distance from mean $x - \bar{x}$	-20	-12	-8	-4	0	4	10	30	0
Squared $(x - \bar{x})^2$	400	144	64	16	0	16	100	900	1,640

- **Range** = $160 - 110 = 50$ mmHg
- **Variance** = $1,640 \div (8 - 1) = 234.3$ mmHg²
- **SD** = $\sqrt{234.3} = 15.3$ mmHg
- **Quartiles:** the lower half is 110, 118, 122, 126, so $Q_1 = (118 + 122) \div 2 = 120$. The upper half is 130, 134, 140, 160, so $Q_3 = (134 + 140) \div 2 = 137$. **IQR** = $137 - 120 = 17$ mmHg.

In a report: "Mean systolic blood pressure was 130 mmHg (SD 15.3)."

Why divide by $n - 1$ and not n ?

The sample mean sits closer to the sample's own values than the true population mean does, so distances measured from it are a little too small. Dividing by $n - 1$ corrects this. The quantity $n - 1$ is called the **degrees of freedom**: once the mean is fixed, only $n - 1$ of the values are free to vary. For a full population (a census) you divide by N .

How outliers affect each measure

Suppose the last patient's reading was 220 instead of 160.

Measure	Original data (max 160)	With outlier (max 220)	Affected?
Mean	130.0	137.5	Yes
Standard deviation	15.3	34.6	Yes, strongly
Range	50	110	Yes, strongly
Median	128	128	No
IQR	17	17	No

REMEMBER

Mean goes with SD. Median goes with IQR. The mean and SD are sensitive to extreme values. The median and IQR are **robust**, which means extreme values barely move them.

Coefficient of variation (CV)

The SD has units, so you cannot directly compare the spread of weight (kg) with the spread of haemoglobin (g/dL). The CV expresses the SD as a percentage of the mean, which removes the unit.

$$CV = \frac{s}{\bar{x}} \times 100\%$$

For the blood pressure data, $CV = 15.3 \div 130 \times 100 = 11.8\%$. For the 2023 monthly dengue counts the SD is 31,933 and the mean is 26,765, so $CV = 119\%$. A CV above 100% says the data vary more than their own average, which is typical of outbreak data. Laboratories also use CV to check whether a test gives consistent results.

1.7 Percentiles, quartiles, the five-number summary and outliers**IN SIMPLE WORDS**

The **pth percentile** is the value below which p% of the data fall. The 90th percentile of birth weight is the weight that 90% of babies are lighter than. **Quartiles** are three special percentiles: Q1 is the 25th, Q2 is the 50th (the median) and Q3 is the 75th.

REAL EXAMPLE: COVID-19 QUARANTINE LENGTH

Real data

Lauer and colleagues (Annals of Internal Medicine, 2020) analysed 181 confirmed cases and estimated the incubation period of COVID-19, the time from infection to first symptoms.

- 2.5th percentile: 2.2 days
- 50th percentile (median): 5.1 days
- 97.5th percentile: 11.5 days

The policy question was "how long must an exposed person be quarantined?" The median cannot answer it, because half of infected people develop symptoms later than 5.1 days. The upper percentiles answer it. The authors estimated that after a 14-day quarantine only about 101 of every 10,000 infected people would develop symptoms later. This is how a percentile became the 14-day rule used worldwide.

The **five-number summary** is minimum, Q_1 , median, Q_3 , maximum. A **box plot** draws it: the box runs from Q_1 to Q_3 , a line marks the median, and "whiskers" extend to the smallest and largest values that are not outliers.

The $1.5 \times \text{IQR}$ rule for outliers

A value is flagged as a possible outlier if it lies below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$. These two limits are called fences.

WORKED EXAMPLE: LENGTH OF STAY AGAIN

Illustrative numbers

Data: 2, 2, 3, 3, 3, 4, 4, 5, 6, 28. Lower half 2, 2, 3, 3, 3 gives $Q_1 = 3$. Upper half 4, 4, 5, 6, 28 gives $Q_3 = 5$. $\text{IQR} = 2$.

Lower fence = $3 - 1.5 \times 2 = 0$. Upper fence = $5 + 1.5 \times 2 = 8$. The 28-day stay is above 8, so it is flagged. Five-number summary: 2, 3, 3.5, 5, 28.

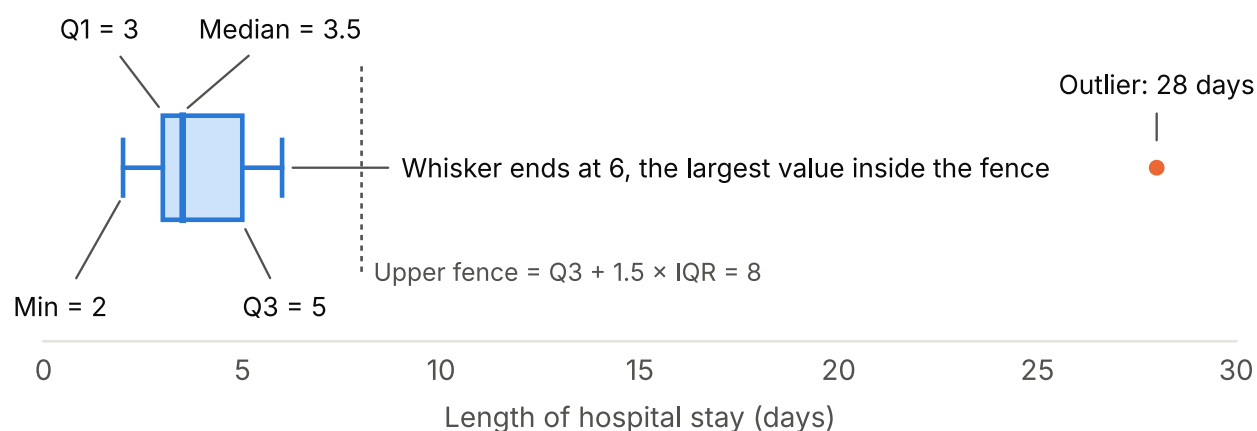


Figure 1.2. How to read a box plot, drawn from the length-of-stay example. The box holds the middle 50% of patients.

COMMON MISTAKE

Deleting outliers automatically. An outlier is a flag to investigate, not an instruction to delete. First ask whether it is a data-entry error (a weight of 650 kg) or a real value (a 28-day stay for a patient with severe dengue). Errors are corrected or removed with a note. Real values stay in the data, and you use robust summaries such as the median. In outbreak detection the "outliers" are the signal you are looking for.

Note on quartiles: there are several accepted ways to calculate quartiles in small samples. These notes split the data at the median and take the median of each half. R, Excel, SPSS and Python may each give slightly different quartiles for small data sets. With large data the differences disappear.

1.8 The shape of a distribution: skewness and kurtosis

IN SIMPLE WORDS

A **distribution** shows which values a variable takes and how often. Draw it as a histogram and look at the shape. **Skewness** measures whether one tail is longer than the other.

Kurtosis measures how heavy the tails are, which means how often extreme values occur.

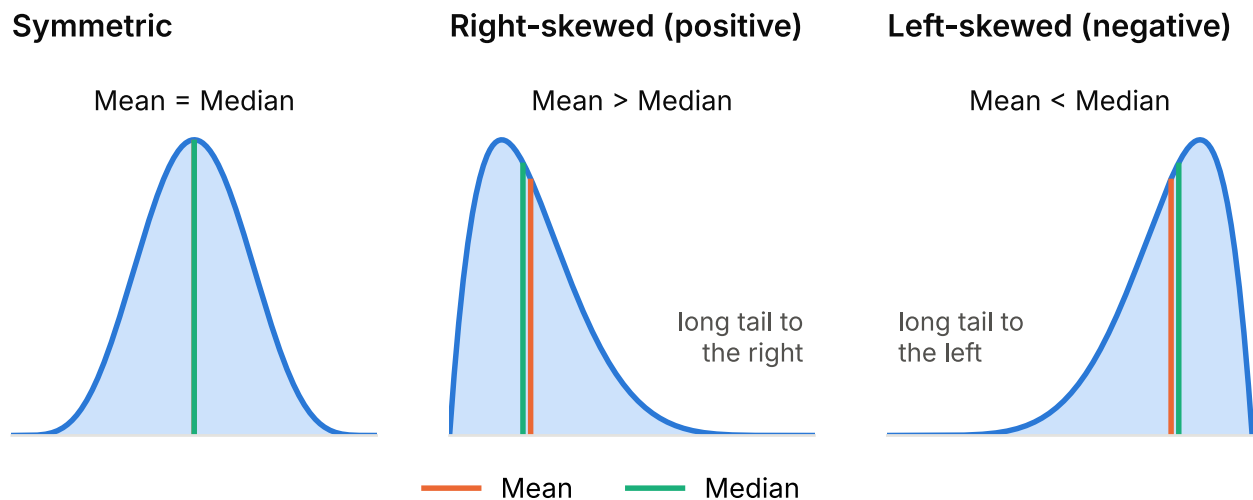


Figure 1.3. Three common shapes. The mean is pulled toward the long tail. The median stays near the bulk of the data.

Shape	Mean and median	Typical public health variables
Symmetric	Mean $\approx$ median	Adult height, haemoglobin, blood pressure in a general population, birth weight of term babies
Right-skewed (positive skew)	Mean > median	Income, health expenditure, length of hospital stay, incubation period, number of clinic visits, antibody titre, parasite count
Left-skewed (negative skew)	Mean < median	Age at death in a high-income country, gestational age at birth

Most health-care data are right-skewed, because values cannot go below zero but can be very large. Other shape words you will meet:

- **Unimodal** means one peak. **Bimodal** means two peaks, which often signals two different groups mixed together, for example an outbreak that hits young children and elderly people but not adults in between.
- **Skewness value:** 0 means symmetric. As a rough guide, between -0.5 and 0.5 is nearly symmetric, and beyond -1 or 1 is strongly skewed.
- **Kurtosis value:** a normal distribution has kurtosis 3. Software usually reports "excess kurtosis", which is kurtosis minus 3, so 0 means normal-like tails and a positive value means heavier tails and more outliers.

REMEMBER

Always draw a histogram before calculating anything. If the mean and median are far apart, the data are skewed: report the median and IQR, and think about a log transformation or a non-parametric test later.

1.9 The z-score: how far from average, in SD units

IN SIMPLE WORDS

A **z-score** tells you how many standard deviations a value is above or below the mean. A z-score of 0 is exactly average. A z-score of -2 is two SDs below average.

$$z = \frac{x - \mu}{\sigma}$$

Converting to z-scores is called **standardising**. It puts different measurements on one common scale, so a child's height and weight can be compared with a reference population of the same age and sex.

REAL EXAMPLE: HOW STUNTING IS DEFINED

Real data

The World Health Organization (WHO) Child Growth Standards give, for every age and sex, the median height of healthy children and the SD. For boys aged 24 months the median length is 87.8 cm and the SD is about 3.06 cm.

A 24-month-old boy who measures 80.5 cm has a height-for-age z-score of

$$z = \frac{80.5 - 87.8}{3.06} = -2.39$$

WHO defines a child as **stunted** when the height-for-age z-score is below -2 , and severely stunted below -3 . This boy is stunted. The same idea gives **wasting** (weight-for-height z-score below -2) and **underweight** (weight-for-age z-score below -2).

In a healthy reference population only about 2.3% of children fall below -2 . BDHS 2022 found 24% of Bangladeshi under-5 children below -2 for height, about ten times the expected share. Wasting was 11% and underweight 22%.

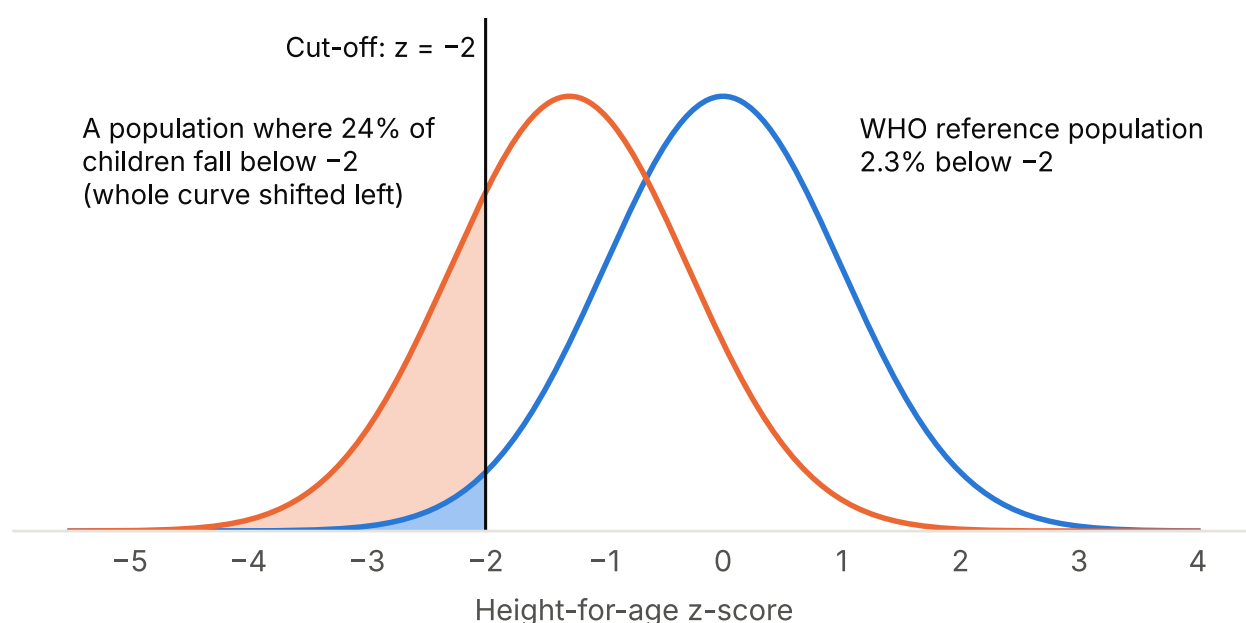


Figure 1.4. Why 24% stunting is a population problem. The orange curve is an illustration of a population in which 24% of children fall below the cut-off: the whole distribution has moved left, not only its tail.

Other uses of z-scores: comparing a laboratory result with a reference range, finding unusual values (a z-score beyond ± 3 is rare), and comparing results measured on different scales.

1.10 Describing the relationship between two variables

Two categorical variables: the cross-tabulation

A cross-tabulation (contingency table) counts people in every combination of two categorical variables. The most important one in public health is the **2 × 2 table** of exposure by outcome. Always calculate percentages within each exposure group (row percentages), so the groups can be compared fairly.

REAL EXAMPLE: THE 1954 SALK POLIO VACCINE TRIAL

Real data

Group	Children	Polio cases	Cases per 100,000
Vaccine	200,745	57	28.4
Placebo	201,229	142	70.6

The counts 57 and 142 look small next to 200,000. The rates per 100,000 make the comparison clear: the placebo group had about 2.5 times the polio rate of the vaccine group. Part 3 tests whether a difference this large could be due to chance.

Two numerical variables: covariance and correlation**IN SIMPLE WORDS**

- **Covariance** shows the direction in which two variables move together. Positive: when one is high the other tends to be high. Negative: when one is high the other tends to be low. Its size depends on the units, so it is hard to interpret.
- **Correlation (Pearson's r)** is covariance with the units removed. It always lies between -1 and $+1$ and measures how closely the points follow a *straight line*.

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1} \quad r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

WORKED EXAMPLE: AGE AND BLOOD PRESSURE

Illustrative numbers

Five adults: ages 30, 40, 50, 60, 70 and systolic blood pressure 118, 124, 130, 140, 148. Mean age is 50 and mean blood pressure is 132.

Age x	30	40	50	60	70	Sum
$x - \bar{x}$	-20	-10	0	10	20	
$y - \bar{y}$	-14	-8	-2	8	16	
Product	280	80	0	80	320	760
$(x - \bar{x})^2$	400	100	0	100	400	1,000
$(y - \bar{y})^2$	196	64	4	64	256	584

Covariance = $760 \div 4 = 190$. Correlation $r = 760 \div \sqrt{1,000 \times 584} = 760 \div 764.2 = \mathbf{0.99}$, a very strong positive relationship. $r^2 = 0.99$ means that 99% of the variation in blood pressure in this small group moves together with age.

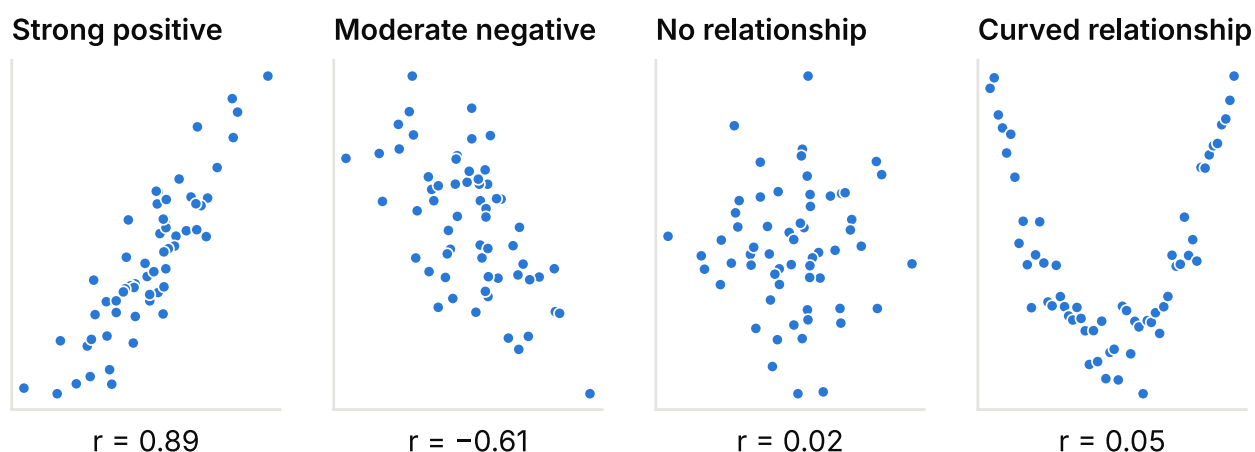


Figure 1.5. What different values of r look like. The last panel is the warning: a strong curved relationship can have r close to zero, because r only measures straight-line relationships. Always look at the scatter plot.

Size of r (ignore the sign)	Rough description	Note
0.00 to 0.19	Very weak	These labels are a common guide only. What counts as "strong" depends on the field. In studies of human behaviour 0.3 can be important. In laboratory calibration 0.9 can be poor.
0.20 to 0.39	Weak	
0.40 to 0.59	Moderate	
0.60 to 0.79	Strong	
0.80 to 1.00	Very strong	

Spearman's rank correlation (r_s or ρ) replaces each value with its rank (1st, 2nd, 3rd...) and then calculates Pearson's r on the ranks. Use it when the data are ordinal, skewed, contain outliers, or follow a curve that keeps rising or keeps falling.

COMMON MISTAKE

Correlation is not causation. Three other explanations must be ruled out before claiming that X causes Y:

- **Confounding:** a third variable drives both. Ice cream sales and drowning deaths rise together because both rise in hot weather.
- **Reverse causation:** Y causes X. Sick people may stop exercising, so "no exercise" and "illness" correlate.
- **Chance:** with many variables some will correlate by accident.

A related error is the **ecological fallacy**: finding a correlation between district averages and then assuming it holds for each individual person.

1.11 Choosing the right chart

What you want to show	Chart	Notes
Counts or percentages across categories	Bar chart	Bars have gaps between them. Start the axis at zero. Sort bars by size unless the categories have a natural order.
Shape of one numerical variable	Histogram	Bars touch because the scale is continuous. Try more than one bin width.
Compare a numerical variable across groups	Box plot	Shows median, spread and outliers for each group side by side.
Cases over time during an outbreak	Epidemic curve	A histogram of cases by date of onset. Its shape suggests how the disease spreads (Figure 1.1).
Trend of a rate over years	Line chart	Time on the horizontal axis.
Relationship between two numerical variables	Scatter plot	Outcome on the vertical axis.
Rates across areas	Shaded map	Map rates, not raw counts, or the map will only show where people live.
Shares of a whole	Bar chart, or pie chart for 2 or 3 groups only	Angles are hard to compare. Avoid pie charts with many slices and all 3D charts.

1.12 Reporting descriptive statistics: "Table 1"

Almost every public health paper begins with a table describing the study participants. The rules:

- Categorical variables: number and percentage, written as n (%).
- Symmetric numerical variables: mean (SD).
- Skewed numerical variables: median (IQR), giving Q1 and Q3.
- State the total sample size, and report how many values are missing for each variable.
- Use sensible rounding. One decimal place is enough for most percentages and means.

EXAMPLE LAYOUT

Illustrative numbers

Characteristic	All participants (n = 400)
Age in years, mean (SD)	41.6 (12.3)
Female, n (%)	216 (54.0)
Residence: urban, n (%)	148 (37.0)
Systolic blood pressure in mmHg, mean (SD)	126.4 (15.8)
Monthly household income in BDT, median (IQR)	22,000 (15,000 to 35,000)
Current smoker, n (%)	88 (22.0)
Body mass index missing, n (%)	12 (3.0)

COMMON MISTAKES IN DESCRIPTIVE STATISTICS

- Reporting a mean without any measure of spread.
- Reporting mean (SD) for skewed data such as income or length of stay. A result like "mean stay 6.0 days (SD 7.8)" is a warning sign: the SD is larger than the mean for a variable that cannot be negative.
- Giving percentages without the count they are based on. "50% improved" could be 1 of 2 patients.
- Confusing SD with standard error (SE). SD describes how spread out the people are. SE describes how precise the mean is. Part 3 explains the difference.
- Calculating a mean of percentages from groups of different sizes without weighting.

1.13 Check yourself: Part 1

1. Household health spending in a survey has mean 4,800 BDT and median 1,900 BDT. What does this tell you, and which should you report?

The mean is much higher than the median, so the data are right-skewed: a few households spent very large amounts. Report the median with the IQR.

2. A district reports 480 tuberculosis cases in a year in a population of 1.2 million. Another reports 150 cases in a population of 250,000. Which has the bigger problem?

Rates: $480 \div 1,200,000 \times 100,000 = 40$ per 100,000, and $150 \div 250,000 \times 100,000 = 60$ per 100,000. The second district has fewer cases but a higher rate.

3. A girl's weight-for-age z-score is -3.2. What does it mean?

Her weight is 3.2 standard deviations below the median of healthy girls of her age. Below -3 is classified as severely underweight.

4. Is "number of antenatal care visits" nominal, ordinal, discrete or continuous?

Discrete. It is a whole-number count.

5. The correlation between a district's number of doctors and its number of deaths is +0.8. Do doctors cause deaths?

No. Population size is a confounder: large districts have more doctors and more deaths.
Compare rates per population instead of counts.

6. Find the median and IQR of these ages at diagnosis: 22, 25, 29, 31, 34, 38, 45, 52.

Median = $(31 + 34) \div 2 = 32.5$. Lower half 22, 25, 29, 31 gives $Q1 = 27$. Upper half 34, 38, 45, 52 gives $Q3 = 41.5$. IQR = 14.5.

Part 2: Probability and Probability Distributions

Probability is the language of uncertainty. Public health uses it every day under other names: risk, prevalence, sensitivity, survival. Distributions are the standard patterns that data follow, and they decide which model you use later.

2.1 What probability means

IN SIMPLE WORDS

Probability is a number between 0 and 1 that says how likely something is. 0 means it cannot happen. 1 means it is certain. 0.24 means it happens about 24 times in every 100.

In public health almost every probability is estimated from data as a **relative frequency**: the number of times the event happened divided by the number of chances it had to happen.

$$P(\text{event}) = \frac{\text{number of people with the event}}{\text{number of people who could have had it}}$$

This is the same as a proportion from Part 1. When the event is a disease, the probability is called **risk**. The basic vocabulary:

Term	Meaning in simple words	Example
Experiment or trial	Any process with an uncertain result	Testing one person for dengue
Outcome	One possible result	Positive
Sample space	The list of all possible outcomes	{Positive, Negative}
Event	The outcome, or set of outcomes, you are interested in	"The test is positive"
Complement , written A^c or "not A"	Everything that is not event A	"The test is not positive"

2.2 The basic rules of probability

All of the rules can be read from one table. The example below is used for the next three sections.

RUNNING EXAMPLE: SMOKING AND HYPERTENSION

Illustrative numbers

A survey of 1,000 adults records smoking status and hypertension.

	Hypertension (H)	No hypertension	Total
Smoker (S)	90	210	300
Non-smoker	140	560	700
Total	230	770	1,000

From the totals: $P(H) = 230/1,000 = 0.23$ and $P(S) = 300/1,000 = 0.30$.

Rule	Formula	In the example
Range	$0 \leq P(A) \leq 1$	All probabilities in the table are between 0 and 1
Complement	$P(\text{not } A) = 1 - P(A)$	$P(\text{no hypertension}) = 1 - 0.23 = 0.77$
Joint probability ("and")	$P(A \text{ and } B)$, written $P(A \cap B)$	$P(H \text{ and } S) = 90/1,000 = 0.09$
Addition rule ("or")	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$	$P(H \text{ or } S) = 0.23 + 0.30 - 0.09 = 0.44$
Addition rule, mutually exclusive events	$P(A \text{ or } B) = P(A) + P(B)$	Used when A and B cannot both happen (see below)

Why subtract in the addition rule? The 90 people who are both smokers and hypertensive are counted once in the 230 and again in the 300. Subtracting removes the double count: $230 + 300 - 90 = 440$ people have at least one of the two.

IN SIMPLE WORDS

Mutually exclusive events cannot happen together. A person's blood group cannot be both A and B at once, so $P(A \text{ or } B)$ is a simple sum. **"Or"** in probability always means "one, or the other, or both".

2.3 Conditional probability and independence

IN SIMPLE WORDS

Conditional probability is the probability of an event *inside a specific group*. $P(A | B)$ is read "the probability of A given B". You stop looking at everybody and look only at the people in group B.

$$P(A | B) = \frac{P(A \text{ and } B)}{P(B)}$$

WORKED EXAMPLE (SAME TABLE) Illustrative numbers

- Risk of hypertension among smokers: $P(H | S) = 90/300 = 0.30$
- Risk of hypertension among non-smokers: $P(H | \text{not } S) = 140/700 = 0.20$
- Share of hypertensive people who smoke: $P(S | H) = 90/230 = 0.39$

The first two are the numbers epidemiology compares. Their ratio, $0.30 \div 0.20 = 1.5$, is the **risk ratio**: in this survey smokers have 1.5 times the risk of hypertension.

COMMON MISTAKE

$P(A | B)$ is not the same as $P(B | A)$. Above, 30% of smokers have hypertension, but 39% of hypertensive people smoke. Swapping the two is the single most common probability error in medicine. The classic case: "most people with lung cancer are smokers" does not mean "most smokers get lung cancer".

Independence

IN SIMPLE WORDS

Two events are **independent** when knowing that one happened tells you nothing about the other. In symbols, $P(A | B) = P(A)$.

In the example $P(H) = 0.23$ but $P(H | S) = 0.30$. Knowing that a person smokes changes the probability of hypertension, so the two are **not independent**. They are associated. Testing whether two variables are independent is exactly what the chi-square test in Part 3 does.

The multiplication rule

Case	Formula
Any two events	$P(A \text{ and } B) = P(A) \times P(B A)$
Independent events	$P(A \text{ and } B) = P(A) \times P(B)$

WORKED EXAMPLE: "AT LEAST ONE" Real data

BDHS 2022 found 24% of under-5 children stunted. Choose 5 children at random from different households, so they are independent.

- $P(\text{one child is not stunted}) = 1 - 0.24 = 0.76$
- $P(\text{none of the 5 is stunted}) = 0.76 \times 0.76 \times 0.76 \times 0.76 \times 0.76 = 0.76^5 = 0.254$
- $P(\text{at least one is stunted}) = 1 - 0.254 = \mathbf{0.746}$

REMEMBER

"At least one" is always easiest through the complement: $P(\text{at least one}) = 1 - P(\text{none}) = 1 - (1 - p)^n$. The same formula explains why running many tests produces false positives: with a 5% false positive chance per test, 10 independent tests on a healthy person give $1 - 0.95^{10} = 40\%$ chance of at least one false alarm.

COMMON MISTAKE

Multiplying probabilities of events that are not independent. Children in the same household share food, water and income, so their stunting status is related. Cases of an infectious disease are not independent either: one case makes the next more likely. Many statistical methods assume independence, and cluster surveys and outbreak data break that assumption.

2.4 Probability and odds

IN SIMPLE WORDS

Odds compare the chance that something happens with the chance that it does not. If 1 person in 5 has a disease, the probability is $1/5 = 0.20$, and the odds are 1 to 4 = 0.25.

$$\text{odds} = \frac{p}{1 - p} \quad p = \frac{\text{odds}}{1 + \text{odds}}$$

Probability p	0.01	0.10	0.20	0.24	0.50	0.80	0.90
Odds $p/(1 - p)$	0.010	0.111	0.25	0.316	1.00	4.00	9.00

Two things to notice. When the probability is small (under about 10%), odds and probability are almost equal. As probability grows, odds become much larger than probability. Odds matter because logistic regression, the main model for yes/no outcomes in public health, works with odds and reports **odds ratios**. For a common outcome, an odds ratio is further from 1 than the risk ratio, so reading an odds ratio as if it were a risk ratio exaggerates the effect.

2.5 Bayes' theorem and diagnostic tests

IN SIMPLE WORDS

Bayes' theorem is the rule for turning $P(B | A)$ into $P(A | B)$. It answers the question a patient actually asks: "My test is positive. What is the chance I really have the disease?"

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)} \quad \text{where} \quad P(B) = P(B | A)P(A) + P(B | \text{not } A)P(\text{not } A)$$

The second formula is the **law of total probability**: the overall probability of B is a weighted average over the groups. In the smoking example, $P(H) = 0.30 \times 0.30 + 0.20 \times 0.70 = 0.09 + 0.14 = 0.23$, which matches the table.

The four numbers that describe a test

	Disease present	Disease absent
Test positive	True positive (TP)	False positive (FP)
Test negative	False negative (FN)	True negative (TN)

Measure	Formula	Question it answers
Sensitivity	$\frac{TP}{TP + FN}$	Of the people who have the disease, what share does the test catch? $P(\text{positive} \mid \text{disease})$
Specificity	$\frac{TN}{TN + FP}$	Of the people who are healthy, what share does the test correctly clear? $P(\text{negative} \mid \text{no disease})$
Positive predictive value (PPV)	$\frac{TP}{TP + FP}$	Of the people who test positive, what share truly have the disease? $P(\text{disease} \mid \text{positive})$
Negative predictive value (NPV)	$\frac{TN}{TN + FN}$	Of the people who test negative, what share are truly healthy? $P(\text{no disease} \mid \text{negative})$

Sensitivity and specificity belong to the test. PPV and NPV depend on the test *and* on how common the disease is in the people being tested. **Prevalence** is the starting probability, also called the prior or pre-test probability.

WORKED EXAMPLE: SCREENING 10,000 PEOPLE Illustrative numbers

A rapid test has sensitivity 90% and specificity 95%. It is used for mass screening in a community where 1% of people have the disease. Count people instead of using the formula. This "natural frequency" method avoids most errors.

- 1 Of 10,000 people, 1% have the disease: 100 diseased and 9,900 healthy.
- 2 Sensitivity 90%: the test catches 90 of the 100 diseased (true positives) and misses 10 (false negatives).
- 3 Specificity 95%: the test correctly clears 9,405 of the 9,900 healthy (true negatives) and wrongly flags 5%, which is 495 (false positives).

	Disease	No disease	Total
Test positive	90	495	585
Test negative	10	9,405	9,415
Total	100	9,900	10,000

PPV = $90 \div 585 = 15.4\%$. **NPV** = $9,405 \div 9,415 = 99.9\%$.

A test that is "90% sensitive and 95% specific" sounds excellent, yet 85 of every 100 positive results here are false. The reason is simple: healthy people are so numerous that even a 5% error among them (495) is far bigger than all the true cases (90).

Now use the same test in a fever clinic during an outbreak, where 30% of patients have the disease: 3,000 diseased give 2,700 true positives, and 7,000 healthy give 350 false positives. $PPV = 2,700 \div 3,050 = 88.5\%$. Same test, same accuracy, very different meaning.

Prevalence in the group tested	0.1%	1%	5%	10%	30%	50%
PPV (sensitivity 90%, specificity 95%)	1.8%	15.4%	48.6%	66.7%	88.5%	94.7%
NPV	99.99%	99.9%	99.4%	98.8%	95.7%	90.5%

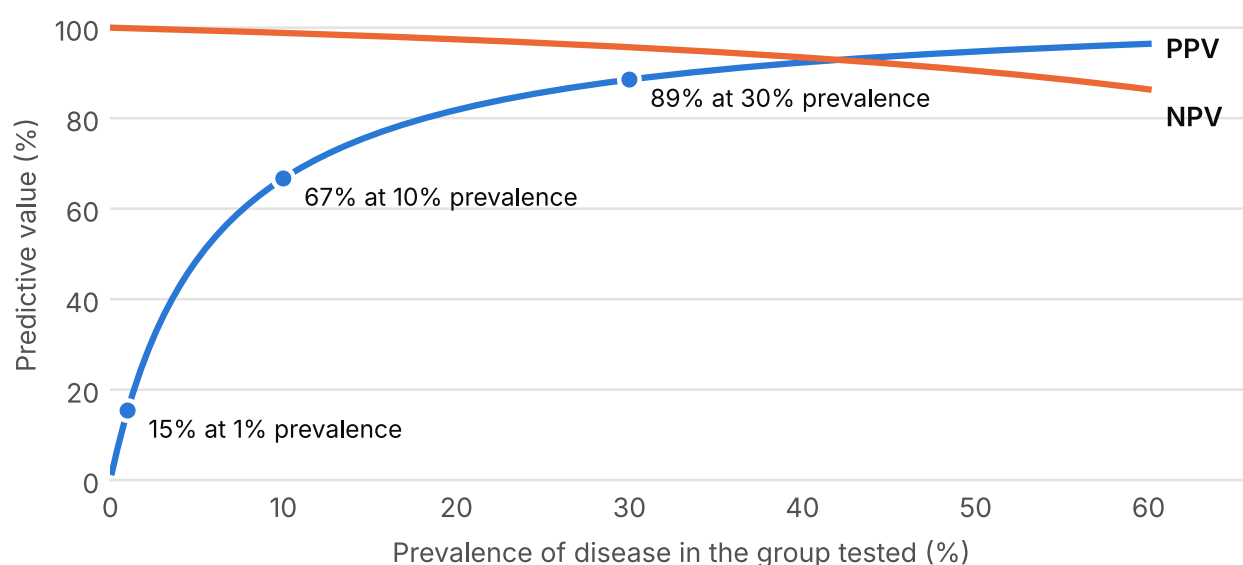


Figure 2.1. Predictive values depend on prevalence (sensitivity 90%, specificity 95%). In low-prevalence screening, a positive result needs a confirmatory test. In high-prevalence settings, a negative result is the one to be careful with.

Likelihood ratios combine sensitivity and specificity into one number that does not depend on prevalence. The positive likelihood ratio is $\text{sensitivity} \div (1 - \text{specificity}) = 0.90 \div 0.05 = 18$: a positive result is 18 times more likely in a diseased person than in a healthy one. The negative likelihood ratio is $(1 - \text{sensitivity}) \div \text{specificity} = 0.10 \div 0.95 = 0.11$.

REMEMBER

- A highly **sensitive** test is good for ruling disease *out* when it is negative. Use it for screening, when missing a case is costly.
- A highly **specific** test is good for ruling disease *in* when it is positive. Use it for confirmation, when a false positive is costly.
- Screening a low-risk population produces mostly false positives, however good the test.

2.6 Random variables, expected value and variance

IN SIMPLE WORDS

A **random variable** is a number whose value depends on chance, such as "number of dengue cases next week" or "birth weight of the next baby". A **discrete** random variable takes separate values you can count (0, 1, 2, ...). A **continuous** random variable can take any value in a range (2.31 kg, 2.317 kg, ...).

Term	Meaning in simple words
Probability distribution	The complete list of values a random variable can take, with the probability of each.
Probability mass function (PMF)	For a discrete variable: the probability of each exact value, $P(X = x)$. The probabilities add to 1.
Probability density function (PDF)	For a continuous variable: a curve. Probability is the <i>area under the curve</i> between two values. The total area is 1. The probability of any single exact value is zero.
Cumulative distribution function (CDF)	The probability of being at or below a value, $P(X \leq x)$. Percentiles are read from the CDF.
Expected value $E(X)$ or μ	The long-run average: what you would get on average if the situation were repeated many times.
Variance $\text{Var}(X)$ or σ^2	The expected squared distance from the expected value. Its square root is the SD.

$$E(X) = \sum x P(X = x) \quad \text{Var}(X) = \sum (x - \mu)^2 P(X = x) = E(X^2) - [E(X)]^2$$

WORKED EXAMPLE: DIARRHOEA CASES PER HOUSEHOLD

Illustrative numbers

Let X be the number of children with diarrhoea in a household in a given week.

x	0	1	2	3	Sum
$P(X = x)$	0.50	0.30	0.15	0.05	1.00
$x \times P$	0	0.30	0.30	0.15	0.75
$x^2 \times P$	0	0.30	0.60	0.45	1.35

$E(X) = 0.75$ cases per household. $\text{Var}(X) = 1.35 - 0.75^2 = 0.79$, so $\text{SD} = 0.89$. No household has 0.75 of a case. The expected value is for planning: 2,000 households would be expected to produce about $2,000 \times 0.75 = 1,500$ cases a week, which tells you how much oral rehydration solution to stock.

2.7 The Bernoulli and binomial distributions: counting "yes" answers

IN SIMPLE WORDS

A **Bernoulli** trial is one event with two possible results, for example one person who is either vaccinated or not. The **binomial distribution** describes how many "yes" results you get when you repeat a Bernoulli trial n times.

Use the binomial when four conditions hold:

- There is a fixed number of trials, n .
- Each trial has only two outcomes (yes or no).
- The probability of "yes", p , is the same for every trial.
- The trials are independent.

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{where} \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$\binom{n}{k}$ is the number of different ways to choose which k of the n people are the "yes" ones. $n!$ (n factorial) means $n \times (n-1) \times \dots \times 1$.

Distribution	Mean	Variance	Standard deviation
Bernoulli(p)	p	$p(1-p)$	$\sqrt{p(1-p)}$
Binomial(n, p)	np	$np(1-p)$	$\sqrt{np(1-p)}$

WORKED EXAMPLE: STUNTED CHILDREN IN A SAMPLE

Real data

With national stunting at 24% (BDHS 2022), a field worker measures 10 randomly chosen under-5 children. $n = 10$, $p = 0.24$.

Probability that exactly 2 are stunted:

$$P(X = 2) = \binom{10}{2} (0.24)^2 (0.76)^8 = 45 \times 0.0576 \times 0.1113 = 0.289$$

Stunted children k	0	1	2	3	4	5 or more
Probability	0.064	0.203	0.289	0.243	0.134	0.067

Expected number = $np = 10 \times 0.24 = 2.4$. SD = $\sqrt{10 \times 0.24 \times 0.76} = 1.35$.

Reading it: finding 0 stunted children in a sample of 10 happens 6.4% of the time even when the true level is 24%. A small sample can easily mislead. And if a village sample of 10 has 5 or more stunted children, that is unusual (6.7%) under the national rate and suggests the village is worse than average.

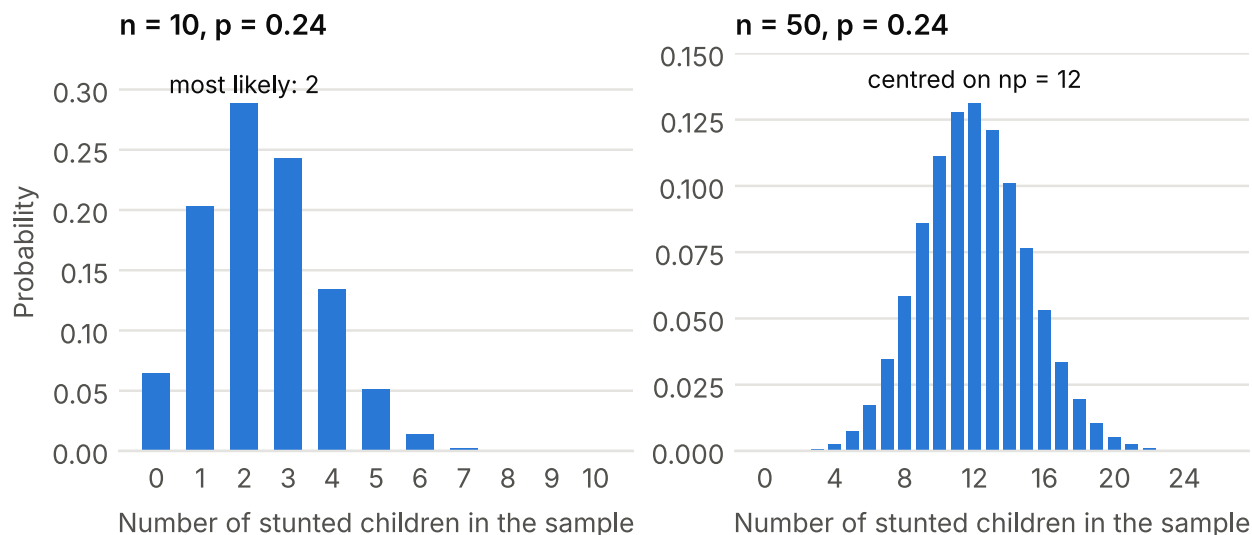


Figure 2.2. Binomial distributions with $p = 0.24$. With 10 children the shape is right-skewed. With 50 children it is close to a symmetric bell centred on 12. This is the basis of the normal approximation in section 2.10.

Where it is used: any count of people with a yes/no characteristic out of a fixed number: positives among those tested, deaths among patients treated, vaccinated children among those surveyed. Every proportion and percentage you report rests on the binomial distribution, and so does logistic regression.

2.8 The Poisson distribution: counting events in time or space

IN SIMPLE WORDS

The **Poisson distribution** describes how many times an event happens in a fixed amount of time, area or population, when events occur independently at a steady average rate. It has one parameter, λ (lambda), the average number of events.

Use the Poisson when you are counting events (0, 1, 2, ...) with no fixed upper limit, the events happen independently of each other, and the average rate is constant over the period.

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad \text{Mean} = \lambda \quad \text{Variance} = \lambda$$

e is the constant 2.718. The key property is that **the mean equals the variance**.

WORKED EXAMPLE: SNAKEBITE ADMISSIONS

Illustrative numbers

A district hospital admits an average of 3 snakebite patients a week, so $\lambda = 3$.

Probability of exactly 5 admissions next week:

$$P(X = 5) = \frac{e^{-3} \times 3^5}{5!} = \frac{0.0498 \times 243}{120} = 0.101$$

Admissions k	0	1	2	3	4	5	6	7 or more
Probability	0.050	0.149	0.224	0.224	0.168	0.101	0.050	0.034

Using it for stock: if the hospital keeps antivenom for 6 patients a week, it runs short only when 7 or more arrive, which is 3.4% of weeks.

Using it for surveillance: 8 or more admissions in one week has probability 1.2% if nothing has changed. A week with 8 admissions is therefore a signal worth investigating (flooding? a change in farming activity?). Outbreak detection systems work on this logic: compare the observed count with what the usual rate predicts.

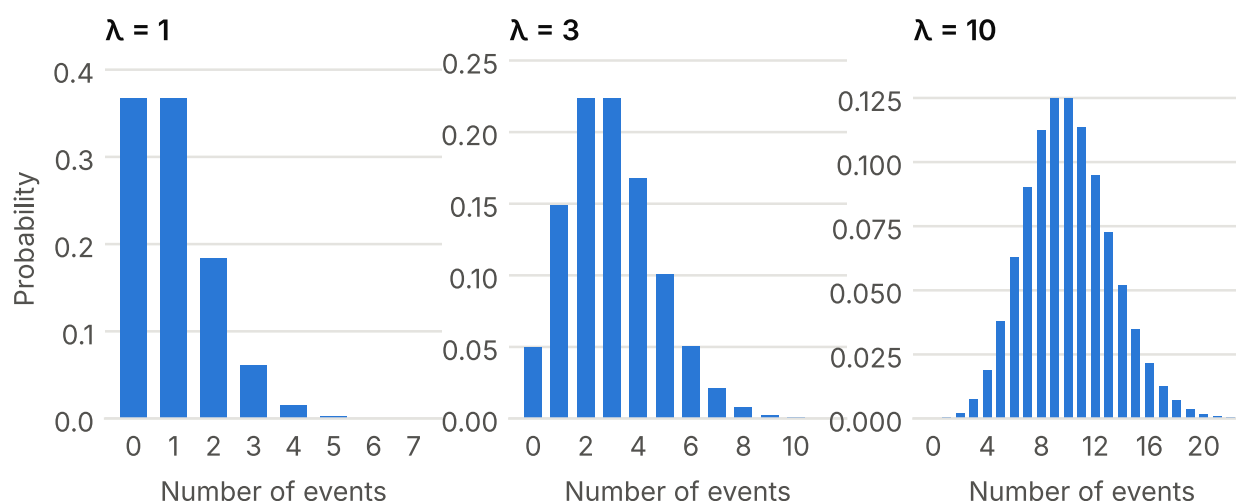


Figure 2.3. Poisson distributions. For rare events (small λ) the distribution is strongly right-skewed and zero is common. As λ grows it becomes symmetric and bell-shaped.

Rates and exposure

Counts depend on how many people were watched and for how long. A Poisson model handles this by writing the expected count as rate $\times$ exposure:

$$\lambda = \text{rate} \times \text{person-time}$$

If the tuberculosis rate is 40 per 100,000 people per year, a town of 250,000 expects $40 \times 2.5 = 100$ cases a year and a town of 25,000 expects 10. In Poisson regression the exposure enters as the "offset". That is a topic for the regression notes, but the idea starts here.

Binomial or Poisson? If each person can only be a yes or a no and you know how many people there were, it is binomial. If you are counting events with no fixed ceiling, or the event is rare in a very large population (n large, p small), use Poisson with $\lambda = np$.

2.9 The negative binomial distribution: counts that vary too much

Real public health counts usually break the Poisson rule that the variance equals the mean. Cases cluster in households, some areas have much higher risk than others, and infectious cases cause further cases. The result is **overdispersion**: the variance is larger than the mean.

IN SIMPLE WORDS

The **negative binomial distribution** is a Poisson distribution with extra spread. It has the same kind of mean, μ , plus a second parameter, k , that controls the extra variation. A small k means strong overdispersion. As k becomes very large, the negative binomial turns back into the Poisson.

$$\text{Mean} = \mu \qquad \text{Variance} = \mu + \frac{\mu^2}{k}$$

REAL EXAMPLE: SUPERSPREADING IN COVID-19 Real data

Early in the pandemic each COVID-19 case infected about 2.5 other people on average. If transmission followed a Poisson distribution, almost every case would infect between 0 and 6 people. That is not what happened. Endo and colleagues (Wellcome Open Research, 2020) fitted a negative binomial distribution and estimated $k = 0.1$ (95% credible interval 0.05 to 0.2) for a mean of 2.5.

With mean 2.5 new infections per case	Poisson	Negative binomial, $k = 0.1$
Variance	2.5	$2.5 + 2.5^2 \div 0.1 = 65$
Cases that infect nobody	8%	72%
Cases that infect 10 or more people	0.03%	7.7%

The study concluded that about 80% of secondary infections may have been caused by roughly 10% of infectious people. The average was the same under both models, but the public health message is completely different: preventing large gatherings and tracing clusters matters far more than treating every case as equally infectious.

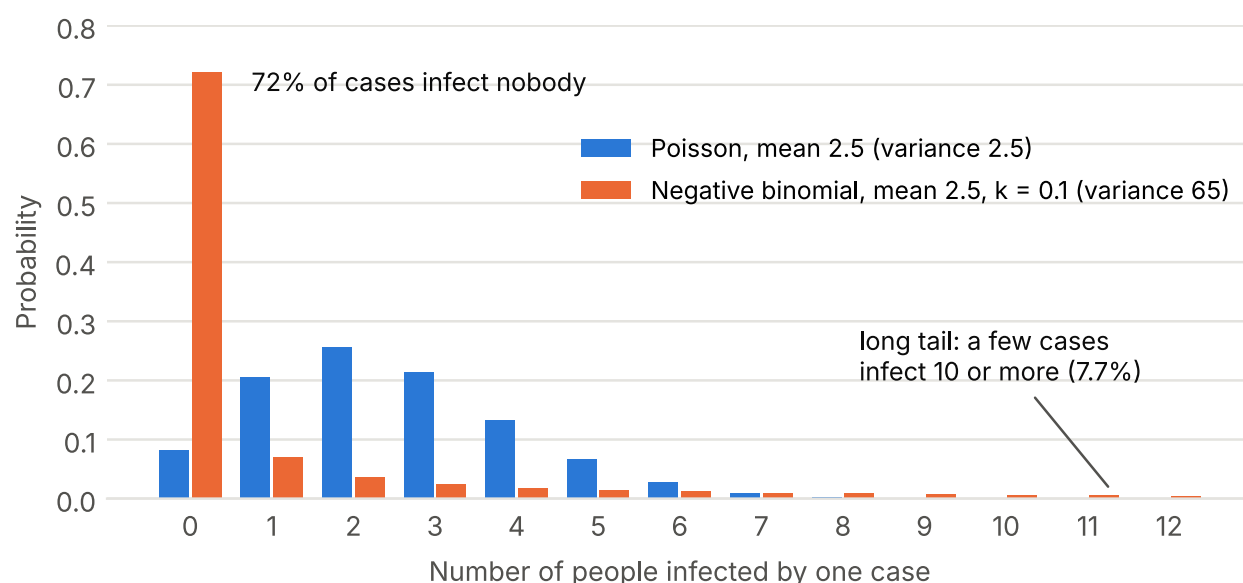


Figure 2.4. Same mean, different story. Both distributions average 2.5 new infections per case. The negative binomial puts most cases at zero and a few far out in the tail.

REMEMBER

For count data, compare the mean with the variance first. If the variance is clearly larger than the mean, the data are overdispersed. A Poisson model will then give standard errors that are too small and p-values that are too optimistic. Use the negative binomial. This is the "Poisson, check overdispersion, negative binomial" path in the roadmap.

2.10 The normal distribution

IN SIMPLE WORDS

The **normal distribution** is the symmetric bell-shaped curve. Most values are near the mean, and values become rarer the further they are from it. It is fully described by two numbers: the mean μ (where the centre is) and the standard deviation σ (how wide it is). It is written $N(\mu, \sigma^2)$.

Properties: symmetric around the mean; mean = median = mode; the curve never quite touches zero; the total area under it is 1.

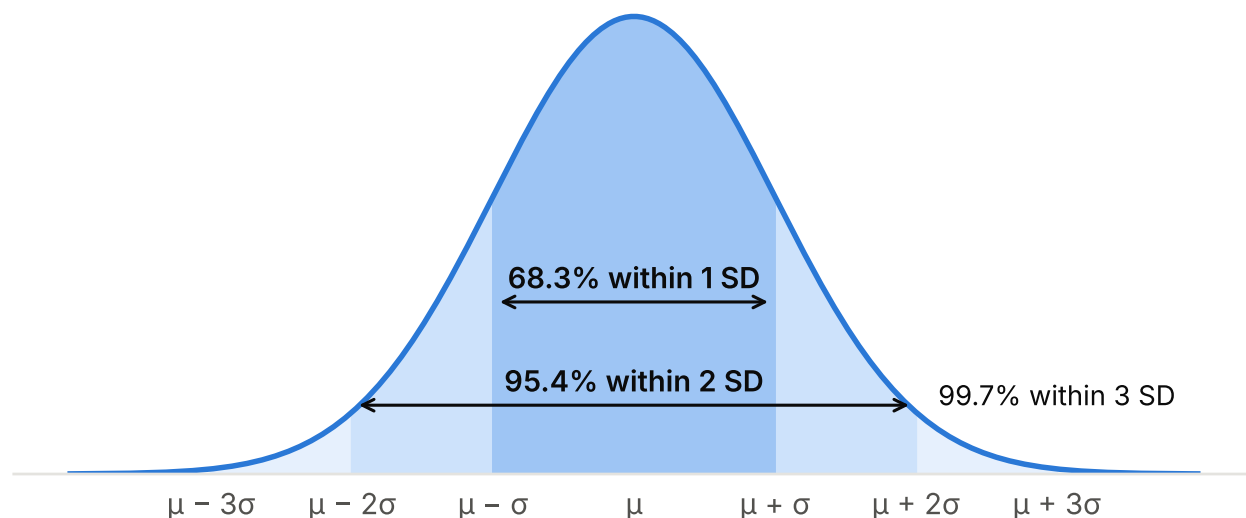


Figure 2.5. The 68-95-99.7 rule. In any normal distribution about 68% of values lie within 1 SD of the mean, 95% within 2 SD (exactly 1.96 SD) and 99.7% within 3 SD.

The **standard normal distribution** has mean 0 and SD 1. Any normal variable can be converted to it with the z-score from Part 1, $z = (x - \mu)/\sigma$, and then one table (the z-table, in the appendix) gives every probability.

z	-1.96	-1.645	-1	0	1	1.28	1.645	1.96	2.576
Area to the left, $P(Z \leq z)$	0.025	0.050	0.159	0.500	0.841	0.900	0.950	0.975	0.995

WORKED EXAMPLE: BLOOD PRESSURE IN A POPULATION

Illustrative numbers

Suppose systolic blood pressure in adults is normal with mean 125 mmHg and SD 15 mmHg.

- **What share have 140 or above?** $z = (140 - 125)/15 = 1.0$. The area to the left of 1.0 is 0.841, so the area to the right is $1 - 0.841 = 15.9\%$.
- **What share lie between 110 and 140?** That is from $z = -1$ to $z = +1$: **68.3%**.
- **What range holds the middle 95%?** $125 \pm 1.96 \times 15 = 95.6$ to **154.4 mmHg**.
- **What is the 90th percentile?** The z-value with 90% to its left is 1.28, so $125 + 1.28 \times 15 = 144.2$ mmHg.

Where it is used in public health:

- **Reference ranges.** A laboratory "normal range" is usually the middle 95% of healthy people, mean ± 1.96 SD. By construction, 5% of healthy people fall outside it.
- **Growth standards.** Stunting, wasting and underweight are defined by z-scores (section 1.9).
- **Statistical inference.** Sample means and sample proportions follow a normal distribution when samples are large, even when the raw data do not. This is the central limit theorem in Part 3, and it is the main reason the normal distribution matters.

Normal approximation to the binomial

When n is large enough that both np and $n(1 - p)$ are at least 10 (some books say 5), a binomial count is approximately normal with mean np and SD $\sqrt{np(1 - p)}$. For a survey of 400 children with $p = 0.24$: mean 96 stunted children, SD 8.5. About 95% of such surveys would find between $96 - 1.96 \times 8.5 = 79$ and $96 + 1.96 \times 8.5 = 113$ stunted children.

COMMON MISTAKE

Assuming data are normal without checking. Many health variables are not: length of stay, cost, counts, titres and waiting times are skewed. Check with a histogram and a Q-Q plot (a plot where normal data fall on a straight line). Also remember that "normal" is only the name of a curve. It does not mean healthy or usual.

2.11 The exponential distribution and other time-to-event distributions

IN SIMPLE WORDS

The **exponential distribution** describes the waiting time until the next event when events happen at a constant average rate. It is the continuous partner of the Poisson: Poisson counts the events, exponential measures the gaps between them.

$$P(T > t) = e^{-\lambda t} \quad \text{Mean waiting time} = \frac{1}{\lambda} \quad \text{Median} = \frac{\ln 2}{\lambda}$$

WORKED EXAMPLE (CONTINUING THE SNAKEBITE EXAMPLE) Illustrative numbers

With 3 admissions a week, the mean gap between admissions is $1/3$ of a week, or 2.3 days. The probability of going a whole week with no admission is $e^{-3} = 0.050$, exactly the Poisson probability of 0 admissions in section 2.8. The median gap is $\ln 2/3$ weeks = 1.6 days, shorter than the mean, because the distribution is right-skewed.

The exponential distribution is the simplest model in **survival analysis**. Its λ is the **hazard**: the instantaneous risk of the event for someone who has not had it yet. The exponential assumes the hazard never changes, which is called being "memoryless". Real hazards change over time, so survival analysis uses more flexible relatives:

Distribution	Simple description	Used for
Weibull	Like the exponential, but the hazard can rise or fall over time	Survival after diagnosis, time to relapse
Gamma	A flexible right-skewed distribution for positive values	Health-care costs, serial intervals, incubation periods
Log-normal	The logarithm of the variable is normal	Incubation periods, antibody titres, exposure concentrations

REAL EXAMPLE Real data

Incubation periods of infectious diseases are right-skewed, and the log-normal distribution has been the standard description since Sartwell's work in 1950. Lauer and colleagues fitted a log-normal model to COVID-19 incubation data in 2020 to obtain the median of 5.1 days and the 97.5th percentile of 11.5 days quoted in section 1.7.

2.12 The distributions behind statistical tests: t, chi-square and F

Three more distributions appear in Part 3. They do not describe raw data. They describe how a *test statistic* behaves when the null hypothesis is true, and they are all built from the normal distribution.

Distribution	Simple description	Used in
Student's t	Looks like the standard normal but with heavier tails. The extra width accounts for estimating the SD from a small sample. Its shape depends on the degrees of freedom (df). Above about 30 df it is almost identical to the normal.	t-tests, confidence intervals for means, regression coefficients
Chi-square (χ^2)	The distribution of a sum of squared standard normal values. Always positive and right-skewed.	Chi-square tests for categorical data, tests of model fit
F	The ratio of two variances. Always positive and right-skewed.	ANOVA, comparing regression models

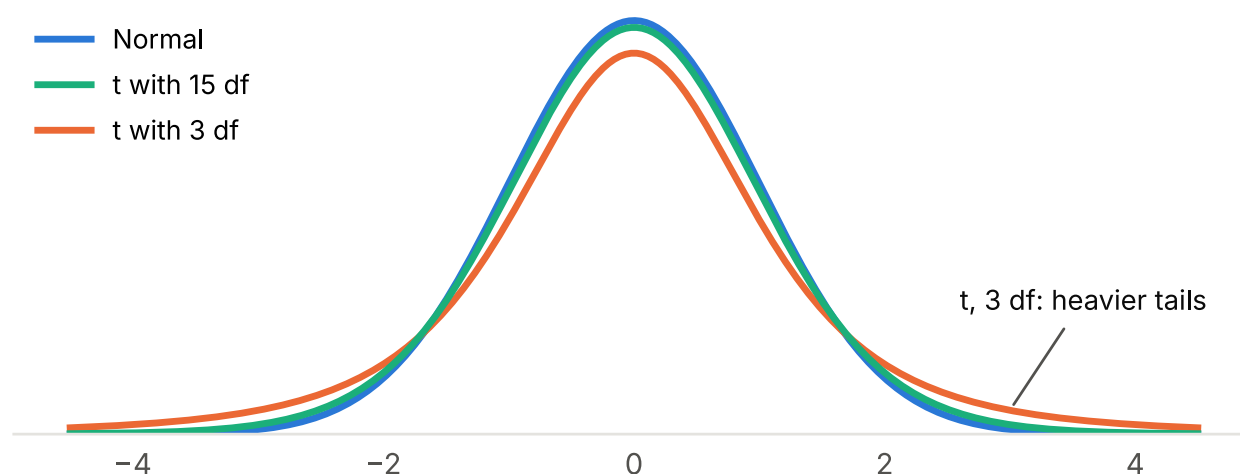


Figure 2.6. The t distribution compared with the normal. With few degrees of freedom the tails are heavier, so the critical value is larger than 1.96 and confidence intervals are wider.

2.13 Choosing a distribution for your data

Your outcome	Distribution	Public health example	Regression model that follows
Yes or no for one person	Bernoulli	Has diabetes or not	Logistic regression
Number of "yes" out of n	Binomial	Positives out of 200 tested	Logistic regression
Count of events, variance about equal to the mean	Poisson	Cases per district per month	Poisson regression
Count of events, variance larger than the mean	Negative binomial	Dengue cases per ward, clinic visits per person	Negative binomial regression
Continuous, symmetric	Normal	Blood pressure, haemoglobin, height	Linear regression
Continuous, positive, right-skewed	Log-normal or gamma	Cost, length of stay, titre	Linear regression on the log, or gamma regression
Time until an event	Exponential, Weibull	Time to death, time to recovery	Survival analysis, Cox regression

COMMON MISTAKES IN PROBABILITY

- **Confusing $P(A | B)$ with $P(B | A)$.** Sensitivity is not PPV. "95% of patients had the risk factor" is not "95% of people with the risk factor become patients".
- **Ignoring the base rate.** Judging a positive test without asking how common the disease is.
- **Assuming independence** for people in the same household, village or outbreak.
- **The gambler's fallacy.** After three quiet weeks an outbreak is not "due". Independent events have no memory.
- **Using the Poisson for overdispersed counts** without checking the variance against the mean.
- **Treating probability 0.05 as impossible.** A 1 in 20 event happens all the time when thousands of districts, weeks or tests are examined.

2.14 Check yourself: Part 2

1. In a village 40% of adults chew betel nut, 15% have oral lesions, and 10% have both. What is the probability that an adult has at least one of the two? What is the probability of oral lesions among chewers?

At least one: $0.40 + 0.15 - 0.10 = 0.45$. Among chewers: $0.10 \div 0.40 = 0.25$, compared with 15% overall, so the two are associated.

2. A test has sensitivity 95% and specificity 90%. Prevalence is 2%. Out of 1,000 people tested, how many positive results are true?

Diseased: 20, of whom 19 test positive. Healthy: 980, of whom 10% = 98 test positive. PPV = $19 \div 117 = 16\%$. About 5 of every 6 positives are false.

3. A health centre sees on average 2 measles cases a month. Which distribution describes the monthly count, and what is the probability of a month with no cases?

Poisson with $\lambda = 2$. $P(0) = e^{-2} = 0.135$.

4. Weekly malaria counts in 50 unions have mean 4 and variance 22. Which distribution fits better, Poisson or negative binomial?

Negative binomial. The variance is more than five times the mean, so the data are overdispersed.

5. Haemoglobin in adult women is normal with mean 12.5 g/dL and SD 1.25. Anaemia is defined as below 12.0. What share are anaemic?

$z = (12.0 - 12.5) \div 1.25 = -0.4$. The area to the left of -0.4 is 0.345, so about 34.5%.

6. If the probability of an outcome is 0.75, what are the odds?

$0.75 \div 0.25 = 3$, or "3 to 1".

Part 3: Inferential Statistics

Inferential statistics uses a sample to say something about the whole population, and attaches an honest measure of uncertainty to it. It has two tools: estimation (confidence intervals) and hypothesis testing (p-values).

3.1 From sample to population

IN SIMPLE WORDS

Inference means drawing a conclusion about a population from a sample. Because a sample is only part of the population, the conclusion always carries uncertainty. Inferential statistics measures that uncertainty.

A sample result can differ from the truth for two very different reasons.

	Sampling error (random error)	Bias (systematic error)
What it is	The luck of the draw. A different random sample would give a slightly different answer.	A flaw in how people were selected or measured that pushes the result in one direction.
Example	One survey of 400 children finds 22% stunted, another finds 26%.	Measuring only children who attend a clinic, who are sicker than average.
Fixed by a bigger sample?	Yes. It shrinks as n grows.	No. A bigger biased sample gives a more precise wrong answer.
Measured by statistics?	Yes: standard errors, confidence intervals, p-values.	No. It must be prevented by good study design.

REMEMBER

Every confidence interval and p-value in this part assumes a **random sample** and measures sampling error only. None of them can detect or correct bias. That is why epidemiology (study design, bias, confounding) is a separate topic in the roadmap.

3.2 Sampling distributions and the standard error

Imagine repeating a survey thousands of times, each time with a new random sample of the same size, and writing down the sample mean each time. Those means would differ a little from each other. The distribution of all those means is called the **sampling distribution** of the mean.

IN SIMPLE WORDS

The **standard error (SE)** is the standard deviation of the sampling distribution. It tells you how much a sample statistic would typically change from one sample to the next. A small SE means a precise estimate.

$$\text{SE of a mean} = \frac{s}{\sqrt{n}} \qquad \text{SE of a proportion} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

	Standard deviation (SD)	Standard error (SE)
Describes	How much individual people differ from each other	How precise the sample mean (or proportion) is
As n increases	Stays about the same	Gets smaller
Use it to	Describe your sample (Table 1)	Build confidence intervals and tests

WORKED EXAMPLE: HAEMOGLOBIN SURVEY

Illustrative numbers

A survey of 400 adolescent girls finds mean haemoglobin 11.2 g/dL with SD 1.6 g/dL.

$$\text{SE} = 1.6 \div \sqrt{400} = 1.6 \div 20 = \mathbf{0.08 \text{ g/dL.}}$$

The SD says individual girls typically differ from the mean by 1.6. The SE says the sample mean of 11.2 would typically move by only 0.08 if the survey were repeated. Because of the square root, quadrupling the sample to 1,600 only halves the SE to 0.04.

3.3 The central limit theorem

IN SIMPLE WORDS

The **central limit theorem (CLT)** says: if you take a large enough random sample, the sample mean follows an approximately normal distribution, *whatever the shape of the original data*. The sampling distribution is centred on the true population mean and its SD is the standard error.

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{approximately, when } n \text{ is large}$$

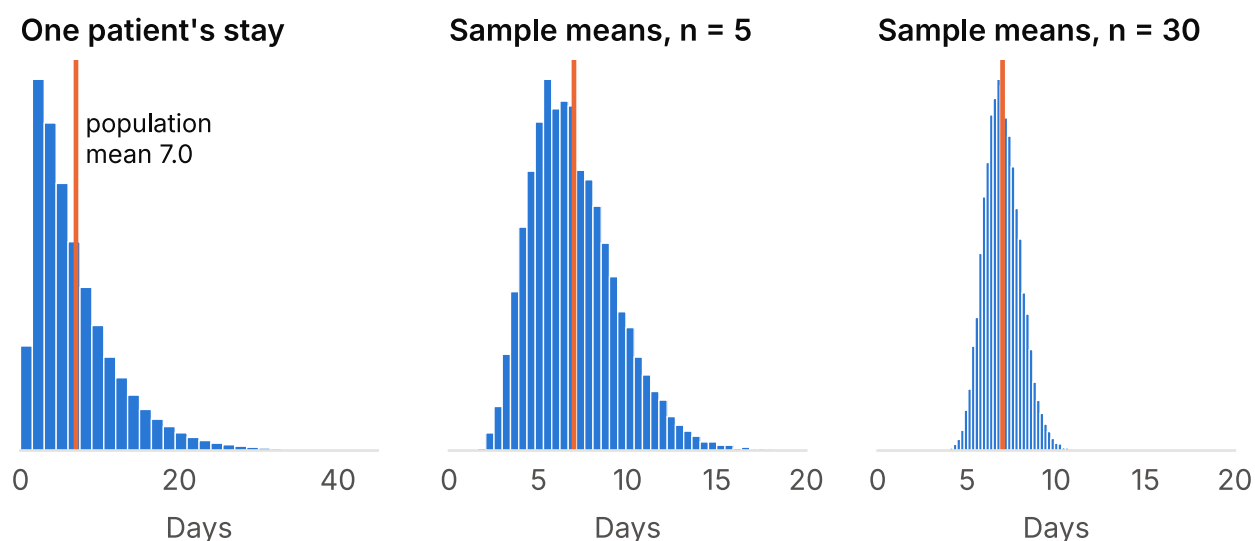


Figure 3.1. The central limit theorem in action (simulated length-of-stay data). Individual stays are strongly right-skewed. Means of samples of 5 are less skewed. Means of samples of 30 are close to a narrow, symmetric bell centred on the population mean.

Why this matters: hospital stays, costs and counts are not normal, but the CLT lets us use normal-based confidence intervals and tests for their *means* and *proportions* as long as the sample is large enough.

How large is "large enough"? The common rule is $n \geq 30$. For strongly skewed data more is needed. For a proportion, the usual check is that the sample contains at least 10 people with the characteristic and at least 10 without it.

The **law of large numbers** is a related idea: as the sample grows, the sample mean gets closer to the true mean. The law of large numbers says the estimate becomes accurate. The CLT describes the shape of its remaining error.

3.4 Estimation: point estimates and confidence intervals

IN SIMPLE WORDS

- A **point estimate** is a single best guess for the population value, such as "24% are stunted".
- A **confidence interval (CI)** is a range of values around the estimate that is likely to contain the true population value. A 95% CI is built by a method that captures the truth in 95 out of 100 repeated samples.

Almost every confidence interval has the same structure:

$$\text{estimate} \pm \underbrace{\text{critical value} \times \text{standard error}}_{\text{margin of error}}$$

Confidence level	90%	95%	99%
Critical value z	1.645	1.96	2.576

Confidence interval for a mean

$$\bar{x} \pm t_{n-1} \times \frac{s}{\sqrt{n}}$$

The critical value comes from the t distribution with $n - 1$ degrees of freedom, because the SD is estimated from the sample. For large samples it is almost 1.96.

WORKED EXAMPLE: HAEMOGLOBIN

Illustrative numbers

- **n = 400**, mean 11.2, SD 1.6, SE 0.08. Critical value 1.97. 95% CI = $11.2 \pm 1.97 \times 0.08 = 11.04$ to 11.36 g/dL.
- **n = 25** with the same mean and SD: SE = $1.6 \div 5 = 0.32$. Critical value for 24 df = 2.064. 95% CI = $11.2 \pm 2.064 \times 0.32 = 10.54$ to 11.86 g/dL.

The small study gives the same estimate but a much wider interval. If the cut-off for anaemia is 12.0 g/dL, both intervals lie entirely below it, so even the small study supports the statement that mean haemoglobin in this group is below the cut-off.

Confidence interval for a proportion

$$\hat{p} \pm z \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

WORKED EXAMPLE: A DISTRICT STUNTING SURVEY

Illustrative numbers

A district survey measures 400 children and finds 96 stunted, so $\hat{p} = 0.24$.

SE = $\sqrt{0.24 \times 0.76/400} = 0.0214$. Margin of error = $1.96 \times 0.0214 = 0.042$. 95% CI = **19.8% to 28.2%**.

Sample size	Standard error	Margin of error	95% confidence interval
100	4.3%	$\pm 8.4\%$	15.6% to 32.4%
400	2.1%	$\pm 4.2\%$	19.8% to 28.2%
1,600	1.1%	$\pm 2.1\%$	21.9% to 26.1%

To halve the margin of error you need four times the sample. Precision is expensive.

This simple formula is called the Wald interval. It works poorly when the sample is small or the proportion is near 0% or 100%. Statistical software then uses the Wilson or exact (Clopper-Pearson) interval. Complex surveys such as BDHS also adjust the SE for clustering and weighting (see the design effect in section 3.8).

What a 95% confidence interval means

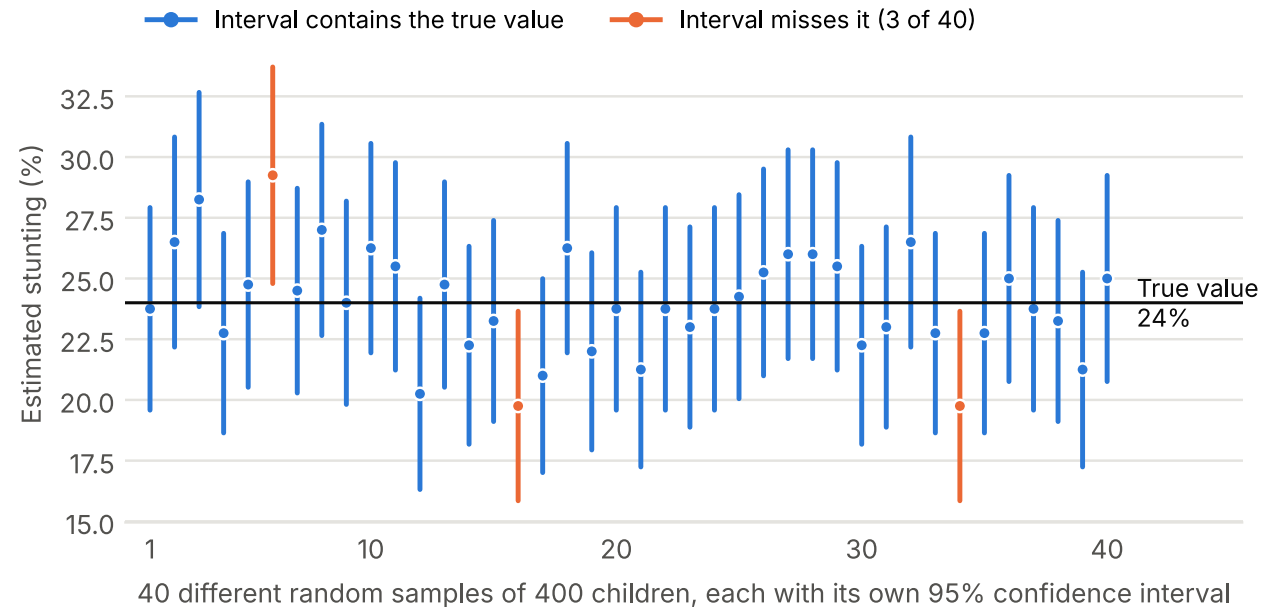


Figure 3.2. Forty surveys of the same population (simulated, true stunting 24%, 400 children each). Each survey gives a different estimate and a different interval. Most intervals contain the true value and a few miss. In the long run 95% contain it. In real life you run one survey and never know whether yours is one of the misses.

Correct	Incorrect
"We are 95% confident that the true stunting prevalence in the district is between 19.8% and 28.2%."	"95% of children have a value between 19.8% and 28.2%." (A CI is about the population average, not about individuals.)
"The method used produces intervals that contain the true value 95% of the time."	"There is a 95% probability that the true value is in this particular interval." (Strictly, the true value is fixed and the interval either contains it or does not. In everyday reporting this wording is common, but know the difference.)
"Values inside the interval are compatible with the data. Values outside are not well supported."	"If we repeat the study, 95% of new estimates will fall inside this interval."

What makes an interval wider or narrower

- **Sample size:** larger n gives a narrower interval.
- **Variability:** a larger SD gives a wider interval.
- **Confidence level:** 99% is wider than 95%, which is wider than 90%. More confidence costs precision.

3.5 Hypothesis testing and the p-value

IN SIMPLE WORDS

A **hypothesis test** asks: "Could the difference I see in my sample be due to chance alone?" It works like a court trial. The starting assumption is "no effect" (innocent). You reject it only if the evidence against it is strong.

Term	Meaning in simple words
Null hypothesis H_0	The "nothing is happening" statement: no difference, no association, no effect. Example: the vaccine and placebo groups have the same polio risk.
Alternative hypothesis H_1 or H_a	What you suspect is true instead: there is a difference.
Test statistic	One number that measures how far the sample result is from what H_0 predicts, in standard-error units. Examples: z , t , χ^2 , F .
p-value	The probability of getting a result at least as extreme as yours <i>if the null hypothesis were true</i> .
Significance level α	The cut-off chosen before the study, usually 0.05. If $p < \alpha$, the result is called "statistically significant".

The five steps

- 1 **State the hypotheses.** Write H_0 and H_1 in terms of population values.
- 2 **Choose the significance level** ($\alpha = 0.05$ unless there is a reason for another value) and the right test for your data (section 3.9).
- 3 **Calculate the test statistic:** (observed value – value expected under H_0) ÷ standard error.
- 4 **Find the p-value** from the matching distribution.
- 5 **Conclude in plain language**, with the effect size and confidence interval, not only "significant" or "not significant".

WORKED EXAMPLE: IS THIS DISTRICT DIFFERENT FROM THE NATIONAL LEVEL?

Illustrative numbers

National stunting is 24% (BDHS 2022). A district survey of 400 children finds 120 stunted, which is 30%.

- 1 H_0 : the district's true prevalence is 0.24. H_1 : it is not 0.24.
- 2 $\alpha = 0.05$. One sample, one proportion, large n : one-sample z-test.
- 3 Under H_0 the SE is $\sqrt{0.24 \times 0.76/400} = 0.0214$. $z = (0.30 - 0.24)/0.0214 = 2.81$.
- 4 The area beyond ± 2.81 in the standard normal is $p = 0.005$.
- 5 If the district were truly at 24%, a sample this far away would occur only 5 times in 1,000. We reject H_0 : stunting in the district is higher than the national level, by an estimated 6 percentage points.

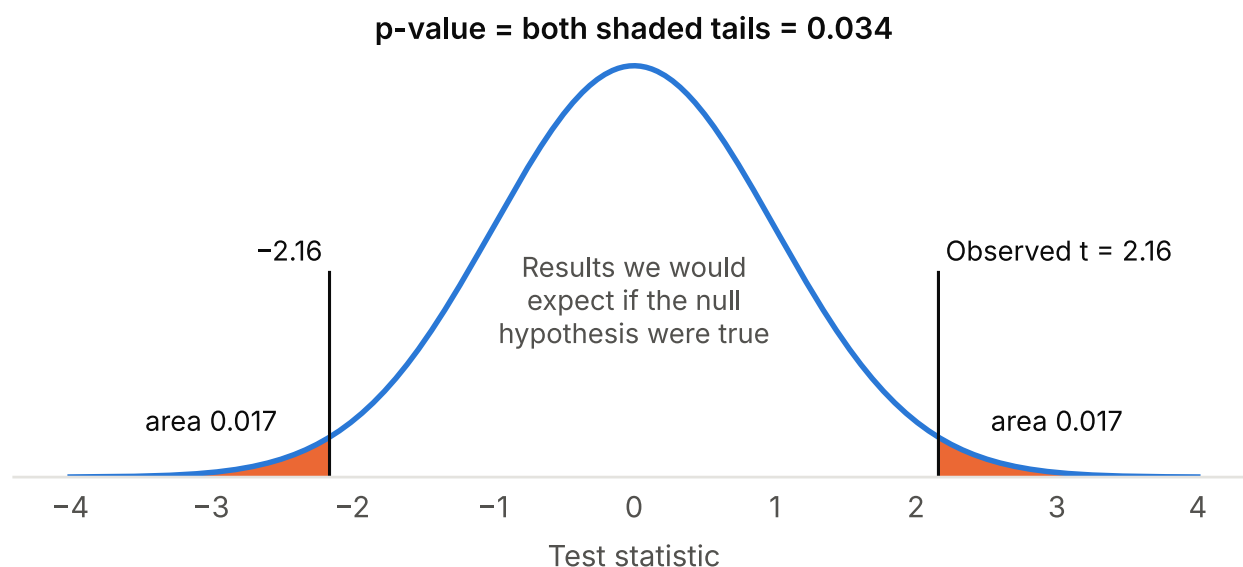


Figure 3.3. What a p-value is. The curve shows the test statistics we would expect if the null hypothesis were true. The p-value is the shaded area: the chance of a statistic at least as far from zero as the one observed. The numbers are from the two-sample example in section 3.10.

What a p-value is not

In 2016 the American Statistical Association published a formal statement because p-values are misused so widely. Its main points, in simple words:

Wrong belief	The truth
"p = 0.03 means there is a 3% chance the null hypothesis is true."	The p-value is calculated <i>assuming</i> H_0 is true. It cannot give the probability that H_0 is true.
"p = 0.03 means there is a 3% chance the result is due to chance."	Same error in different words.
"A small p-value means a large or important effect."	The p-value mixes effect size with sample size. A tiny, useless effect can have a very small p-value in a big study (section 3.7).
"p > 0.05 means there is no effect."	It means the study did not find convincing evidence. The effect may be real but the sample too small. Absence of evidence is not evidence of absence.
"p = 0.049 is a finding and p = 0.051 is not."	The 0.05 line is a convention. The two results carry almost the same evidence.

One-sided and two-sided tests

A **two-sided** test asks "is there a difference in either direction?" and counts both tails. A **one-sided** test asks only about one direction and counts one tail, which halves the p-value. Use two-sided tests by default. A one-sided test is acceptable only when a difference in the other direction is impossible or irrelevant, and it must be decided before seeing the data.

Confidence intervals and tests agree

If a 95% CI for a difference does not contain 0, the two-sided test gives $p < 0.05$. For a ratio (risk ratio, odds ratio) the "no effect" value is 1 instead of 0. The confidence interval tells you more than the p-value, because it also shows how big the effect could be. Report both.

3.6 Type I error, Type II error and power

A test can be wrong in two ways.

	Truth: no real effect (H_0 true)	Truth: a real effect exists (H_0 false)
Test says "significant" (reject H_0)	Type I error: false alarm. Probability = α	Correct. Probability = power = $1 - \beta$
Test says "not significant" (do not reject H_0)	Correct. Probability = $1 - \alpha$	Type II error: missed effect. Probability = β

IN SIMPLE WORDS

- **Type I error (α):** saying there is an effect when there is none. Like a false positive on a diagnostic test.
- **Type II error (β):** missing an effect that is really there. Like a false negative.
- **Power ($1 - \beta$):** the chance that the study detects an effect that truly exists. Like sensitivity. Studies are usually designed for 80% or 90% power.

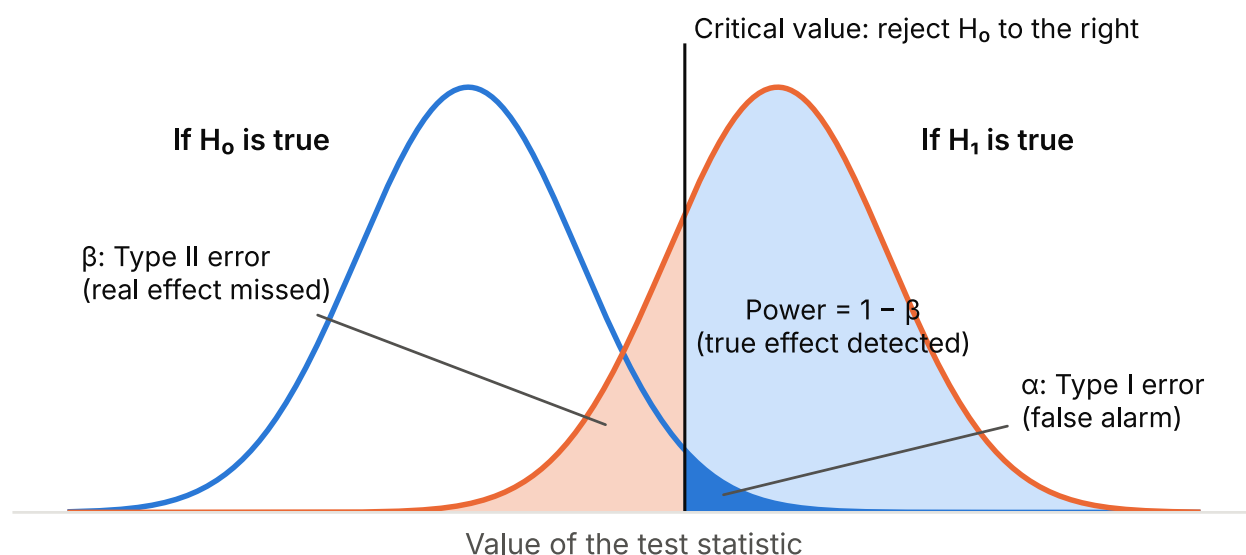


Figure 3.4. The two errors and power. The blue curve is what the test statistic does when there is no effect. The orange curve is what it does when a real effect exists. Moving the critical value to the right reduces false alarms (α) but increases missed effects (β).

Public health consequences. A Type I error can lead to a useless programme being funded or a harmless food being blamed for an outbreak. A Type II error can lead to an effective intervention being abandoned or a real hazard being declared safe. Which error is worse depends on the situation.

Power increases when:

- the sample size is larger;
- the true effect is larger;
- the data are less variable (smaller SD, better measurement);
- α is set higher (0.10 instead of 0.05), which trades more false alarms for fewer missed effects.

WORKED EXAMPLE: HOW SAMPLE SIZE DRIVES POWER

Illustrative numbers

A programme is expected to reduce anaemia from 30% to 20%. The table shows the chance that a study comparing two groups would find a significant difference (two-sided $\alpha = 0.05$), if that reduction is real.

Participants per group	50	100	200	291	400	600
Power	21%	38%	64%	80%	91%	98%

With 50 per group the study would miss this large and important effect four times out of five. A "not significant" result from such a study says almost nothing. This is why sample size must be calculated before data collection.

3.7 Effect size: statistical significance and practical importance

IN SIMPLE WORDS

An **effect size** says *how big* a difference or association is. The p-value only says whether the difference can be told apart from zero. Public health decisions need the size.

Comparison	Effect size	Meaning
Two means	Mean difference, with CI	In the original units, such as mmHg or g/dL. The most useful for decisions.
Two means, standardised	Cohen's d = mean difference $\div$ SD	Common guide: 0.2 small, 0.5 medium, 0.8 large.
Two proportions	Risk difference, risk ratio, odds ratio	Covered in section 3.12 and in the epidemiology notes.
Two numerical variables	Correlation r , or r^2	Strength of the straight-line relationship.
Three or more means	Eta squared (η^2)	Share of the total variation explained by the groups.

WORKED EXAMPLE: SIGNIFICANT BUT UNIMPORTANT

Illustrative numbers

A study with 100,000 people in each group finds that a new salt-awareness leaflet lowers mean systolic blood pressure by 0.3 mmHg (SD 15).

SE of the difference = $15 \times \sqrt{2/100,000} = 0.067$. $z = 0.3/0.067 = 4.47$, giving $p < 0.00001$.

Cohen's $d = 0.3/15 = 0.02$.

The result is "highly significant", yet 0.3 mmHg has no clinical meaning. With a huge sample, almost any difference becomes statistically significant. The opposite also happens: a 6 mmHg reduction in a study of 20 people may be "not significant" and still be worth a larger trial.

REMEMBER

Statistical significance answers "is it real?" Effect size answers "is it big enough to matter?" A confidence interval answers both at once. Always ask all three.

3.8 Sample size basics

Sample size is decided before the study, from what you want to achieve. Too few people and the study cannot answer the question. Too many and money and participants' time are wasted.

To estimate one proportion (a prevalence survey)

$$n = \frac{z^2 p(1-p)}{d^2}$$

Here z is 1.96 for 95% confidence, p is the expected prevalence, and d is the margin of error you can accept, written as a proportion (0.05 for ± 5 percentage points). This is often called Cochran's formula. If you have no idea of p , use 0.5, which gives the largest sample and is therefore the safe choice.

WORKED EXAMPLE

Illustrative numbers

Unknown prevalence, $\pm 5\%$ margin: $n = 1.96^2 \times 0.5 \times 0.5 / 0.05^2 = 3.8416 \times 0.25 / 0.0025 = 384.2$, rounded up to **385**. This is where the "384" seen in so many theses comes from.

Expected prevalence 24%, $\pm 5\%$ margin: $n = 3.8416 \times 0.24 \times 0.76 / 0.0025 = 280.3$, rounded up to **281**.

Margin of error ($p = 0.5$)	$\pm 10\%$	$\pm 5\%$	$\pm 3\%$	$\pm 2\%$	$\pm 1\%$
Sample size needed	97	385	1,068	2,401	9,604

Three adjustments are made in practice:

- **Design effect (DEFF).** Cluster surveys select villages or wards first and then households, because that is cheaper. People in the same cluster are alike, so each one adds less new information. Multiply n by the design effect, often taken as 1.5 to 2. With DEFF 1.5: $385 \times 1.5 = 578$.
- **Non-response.** Divide by the expected response rate. With 10% non-response: $578 \div 0.9 = 642$.
- **Small populations.** If the population N is small, use the finite population correction $n_{\text{adj}} = n / (1 + (n - 1) / N)$. For $N = 2,000$: $384.2 \div (1 + 383.2 / 2,000) = 323$.

To compare two proportions

$$n \text{ per group} = \frac{(z_{\alpha/2} + z_{\beta})^2 [p_1(1 - p_1) + p_2(1 - p_2)]}{(p_1 - p_2)^2}$$

For $\alpha = 0.05$, $z_{\alpha/2} = 1.96$. For 80% power $z_{\beta} = 0.84$, and for 90% power $z_{\beta} = 1.28$.

WORKED EXAMPLE Illustrative numbers

To detect a fall in anaemia from 30% to 20% with 80% power: $(1.96 + 0.84)^2 = 7.85$. The variance part is $0.30 \times 0.70 + 0.20 \times 0.80 = 0.37$. The difference squared is $0.10^2 = 0.01$.

$n = 7.85 \times 0.37 / 0.01 = 290.4$, so **291 per group**, matching the power table in section 3.6.

REMEMBER

The smaller the difference you want to detect, the larger the sample. Halving the difference needs about four times as many people. A sample size statement must list its inputs: confidence level, power, expected values, margin or difference, design effect, and non-response.

3.9 Choosing the right test

Ask three questions. What type is the outcome variable? How many groups are being compared? Are the groups independent, or are the same people measured more than once (paired)?

Outcome	Comparison	Test	If assumptions fail
Numerical	One sample against a known value	One-sample t-test	Wilcoxon signed-rank test
	Two independent groups	Independent t-test (Welch)	Mann-Whitney U test
	Two paired measurements	Paired t-test	Wilcoxon signed-rank test
	Three or more independent groups	One-way ANOVA	Kruskal-Wallis test
	Three or more repeated measurements	Repeated-measures ANOVA	Friedman test
Categorical	One sample against a known proportion	One-sample z-test for a proportion	Exact binomial test
	Two or more independent groups	Chi-square test	Fisher's exact test
	Two paired yes/no measurements	McNemar's test	Exact McNemar test
	Ordered groups (dose levels)	Chi-square test for trend	
Two numerical variables	Strength of relationship	Pearson correlation	Spearman correlation
Binary outcome, several predictors	Adjusting for confounders	Logistic regression	
Count outcome	Rates across groups	Poisson regression	Negative binomial regression

Parametric tests (t-test, ANOVA, Pearson) assume the data, or the sample means, follow a known distribution, usually the normal. **Non-parametric** tests (Mann-Whitney, Wilcoxon, Kruskal-Wallis, Spearman) work on ranks and make fewer assumptions. They are safer for small, skewed or ordinal data, at the cost of slightly less power when the data really are normal.

3.10 Tests for numerical outcomes

Independent two-sample t-test

Question: is the mean of a numerical variable different between two separate groups? H_0 : the two population means are equal.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

This version is **Welch's t-test**, which does not assume the two groups have the same variance. It is the default in R and is the safer choice. The older Student's version pools the two variances.

WORKED EXAMPLE: IRON SUPPLEMENTATION

Illustrative numbers

After 12 weeks, 40 girls who received iron tablets have mean haemoglobin 11.8 g/dL (SD 1.4). Forty girls in the comparison group have mean 11.1 g/dL (SD 1.5).

- Difference = 11.8 – 11.1 = 0.7 g/dL
- SE = $\sqrt{1.4^2/40 + 1.5^2/40} = \sqrt{0.049 + 0.056} = 0.324$
- $t = 0.7/0.324 = 2.16$, with about 78 degrees of freedom, $p = 0.034$
- 95% CI for the difference = $0.7 \pm 1.99 \times 0.324 = \mathbf{0.05 \text{ to } 1.35 \text{ g/dL}}$
- Cohen's $d = 0.7/1.45 = 0.48$, a medium effect

Reporting sentence: "Mean haemoglobin was 0.7 g/dL higher in the supplemented group (95% CI 0.05 to 1.35; $p = 0.034$).\" The interval shows the honest picture: the true benefit could be as small as 0.05 or as large as 1.35.

Paired t-test

Question: did a measurement change in the same people (before and after), or differ between matched pairs? Calculate the difference for each pair, then test whether the mean difference is zero.

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \quad \text{with } n - 1 \text{ degrees of freedom}$$

WORKED EXAMPLE: SALT REDUCTION AND BLOOD PRESSURE

Illustrative numbers

Ten hypertensive patients have systolic blood pressure measured before and after 8 weeks of a low-salt diet.

Patient	1	2	3	4	5	6	7	8	9	10	Mean
Before	150	148	160	155	142	158	165	149	152	157	153.6
After	144	146	151	150	143	149	158	145	147	150	148.3
Difference	6	2	9	5	-1	9	7	4	5	7	5.3

- Mean difference $\bar{d} = 5.3$ mmHg. SD of the differences $s_d = 3.09$.
- $SE = 3.09 \div \sqrt{10} = 0.978$. $t = 5.3/0.978 = 5.42$ with 9 df, $p = 0.0004$.
- 95% CI = $5.3 \pm 2.262 \times 0.978 = 3.1$ to 7.5 mmHg.

Why pairing matters: if these data are wrongly analysed as two independent groups, the result is $p = 0.053$, "not significant". Patients differ a lot from each other (142 to 165), and that variation hides the effect. Pairing removes it, because each patient is compared with himself or herself.

COMMON MISTAKE

A before-and-after study with no control group cannot prove that the diet caused the fall. Blood pressure may drop anyway on a second visit because patients are calmer, or because extreme first readings tend to move toward the average (regression to the mean). The test shows the change is real. Only the study design can show what caused it.

One-way ANOVA (analysis of variance)

Question: are the means of three or more groups all equal? H_0 : all the population means are equal. H_1 : at least one is different.

IN SIMPLE WORDS

ANOVA compares two kinds of variation: how far the group means are from each other (**between-group**) and how spread out people are inside each group (**within-group**). If the groups differ more than the noise inside them can explain, the F statistic is large.

$$F = \frac{\text{variation between groups}}{\text{variation within groups}} = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

WORKED EXAMPLE: HAEMOGLOBIN IN THREE DISTRICTS

Illustrative numbers

Twenty women are sampled in each of three districts. Mean haemoglobin is 11.0, 11.6 and 12.1 g/dL, and the SD within each district is 1.2.

- Overall mean = $(11.0 + 11.6 + 12.1) \div 3 = 11.57$
- Between-group sum of squares = $20 \times [(11.0 - 11.57)^2 + (11.6 - 11.57)^2 + (12.1 - 11.57)^2] = 12.13$, with $3 - 1 = 2$ df, so $MS_{\text{between}} = 6.07$
- $MS_{\text{within}} = 1.2^2 = 1.44$, with $60 - 3 = 57$ df
- $F = 6.07/1.44 = 4.21$, $p = 0.020$

At least one district differs. ANOVA does not say which. Eta squared = $12.13 \div (12.13 + 82.08) = 0.13$: district explains 13% of the variation in haemoglobin.

To find which pairs differ, run a **post hoc test** such as Tukey's HSD or Bonferroni-corrected t-tests. Do not run many separate t-tests without correction (section 3.13).

Non-parametric alternatives

These tests replace the values with their ranks, so an extreme value counts only as "the largest" however large it is.

WORKED EXAMPLE: MANN-WHITNEY U TEST

Illustrative numbers

Length of stay in days for 5 patients on a standard protocol: 7, 9, 11, 14, 30. For 5 patients on a new protocol: 2, 3, 5, 6, 8.

Rank all 10 values from smallest to largest. The new-protocol patients get ranks 1, 2, 3, 4 and 6, which sum to 16. The standard-protocol patients get ranks 5, 7, 8, 9 and 10, which sum to 39.

$U = 16 - \frac{5 \times 6}{2} = 1$. The exact two-sided p-value is **0.016**. Median stay is 5 days against 11 days.

A t-test on the same data gives $p = 0.084$, because the single 30-day stay inflates the variance. With small, skewed samples the rank test is both safer and, here, more powerful.

Non-parametric test	Replaces	Notes
Mann-Whitney U (Wilcoxon rank-sum)	Independent t-test	Report medians and IQRs with it
Wilcoxon signed-rank	Paired t-test	Uses the ranks of the paired differences
Kruskal-Wallis	One-way ANOVA	Follow with Dunn's test for pairwise comparisons
Spearman rank correlation	Pearson correlation	For ordinal, skewed or curved-but-rising data

Testing a correlation

To test whether a correlation differs from zero: $t = r\sqrt{n-2} / \sqrt{1-r^2}$ with $n-2$ df. For the age and blood pressure example in section 1.10 ($r = 0.995$, $n = 5$): $t = 16.5$, $p = 0.0005$. With large samples even $r = 0.1$ becomes "significant", so always judge the size of r , not only its p-value.

3.11 Tests for categorical outcomes

Chi-square test of independence

Question: are two categorical variables related? H_0 : they are independent, meaning the outcome percentage is the same in every group.

IN SIMPLE WORDS

The chi-square test compares the counts you **observed** (O) with the counts you would **expect** (E) if there were no relationship at all. The bigger the gap between observed and expected, the bigger the statistic.

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad E = \frac{\text{row total} \times \text{column total}}{\text{grand total}} \quad df = (\text{rows} - 1)(\text{columns} - 1)$$

REAL EXAMPLE: THE 1954 SALK POLIO VACCINE FIELD TRIAL

Real data

In the randomised, placebo-controlled part of the trial, 401,974 children received either the vaccine or a placebo injection. Neither the children nor the doctors knew who received which.

Observed	Polio	No polio	Total
Vaccine	57	200,688	200,745
Placebo	142	201,087	201,229
Total	199	401,775	401,974

- Expected counts if the vaccine did nothing.** The 199 cases would be shared in proportion to group size. Expected cases in the vaccine group = $200,745 \times 199 \div 401,974 = 99.4$. In the placebo group = $201,229 \times 199 \div 401,974 = 99.6$.
- Compare.** The vaccine group had 57 cases where 99.4 were expected. The placebo group had 142 where 99.6 were expected.
- Statistic.** $(57 - 99.4)^2 \div 99.4 = 18.07$ and $(142 - 99.6)^2 \div 99.6 = 18.03$. The two "no polio" cells add only 0.009 each. $\chi^2 = 36.1$ with 1 df.
- p-value.** The critical value for 1 df at $\alpha = 0.05$ is 3.84. A value of 36.1 gives $p < 0.000001$.

A difference this large would essentially never arise by chance. Combined with randomisation and blinding, which rule out bias and confounding, the trial showed that the vaccine prevented polio. It was licensed in 1955.

Assumptions: observations are independent (each person appears in one cell only), and expected counts are large enough: the usual rule is that every expected count should be at least 5.

Degrees of freedom	1	2	3	4	5	6
Critical value of χ^2 at $\alpha = 0.05$	3.84	5.99	7.81	9.49	11.07	12.59

Fisher's exact test

When any expected count is below 5, the chi-square approximation is unreliable. Fisher's exact test calculates the exact probability of the observed table, and of every more extreme table, using the hypergeometric distribution. It is the standard choice for small 2×2 tables, which are common in outbreak investigations.

WORKED EXAMPLE: A FOOD-POISONING OUTBREAK Illustrative numbers

Twenty guests at a wedding are interviewed. Of the 11 who ate the beef curry, 9 became ill. Of the 9 who did not, 2 became ill.

	Ill	Not ill	Total	Attack rate
Ate beef curry	9	2	11	81.8%
Did not eat it	2	7	9	22.2%

Three of the four expected counts are below 5 (4.95, 4.95 and 4.05), so use Fisher's exact test: **$p = 0.022$** . The risk ratio is $81.8 \div 22.2 = 3.7$. Guests who ate the curry were about 3.7 times as likely to become ill, and this is unlikely to be chance. The curry becomes the main suspect for laboratory testing.

McNemar's test for paired categorical data

Used when the same people are classified twice (before and after, or by two different tests). Only the pairs that *changed* carry information. If b people changed from yes to no and c changed from no to yes:

$$\chi^2 = \frac{(b - c)^2}{b + c} \quad \text{with 1 df}$$

Example with illustrative numbers: 100 patients are tested with two rapid tests. Fifteen are positive on test A only and 5 on test B only. $\chi^2 = (15 - 5)^2 / 20 = 5.0$, $p = 0.025$. Test A gives positive results more often.

3.12 Confidence intervals for risk ratios and odds ratios

Public health papers rarely stop at " $p < 0.05$ " for a 2×2 table. They report how many times higher the risk is, with a confidence interval. Ratios are not symmetric (they cannot go below 0 but can be very large), so the interval is calculated on the log scale and then converted back.

	Outcome	No outcome	Total
Exposed	a	b	n_1
Unexposed	c	d	n_0

$$RR = \frac{a/n_1}{c/n_0} \quad SE(\ln RR) = \sqrt{\frac{1}{a} - \frac{1}{n_1} + \frac{1}{c} - \frac{1}{n_0}} \quad 95\% \text{ CI} = e^{\ln RR \pm 1.96 \times SE}$$

$$OR = \frac{a \times d}{b \times c} \quad SE(\ln OR) = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$

REAL EXAMPLE: THE SALK TRIAL, CONTINUED

Real data

- Risk in the vaccine group = $57 \div 200,745 = 28.4$ per 100,000. Risk in the placebo group = $142 \div 201,229 = 70.6$ per 100,000.
- **Risk ratio** = $28.4 \div 70.6 = \mathbf{0.40}$. Vaccinated children had 40% of the risk of unvaccinated children.
- $\ln(0.40) = -0.910$. $SE = \sqrt{1/57 - 1/200,745 + 1/142 - 1/201,229} = 0.157$.
- 95% CI: $e^{-0.910 - 1.96 \times 0.157}$ to $e^{-0.910 + 1.96 \times 0.157} = \mathbf{0.30 \text{ to } 0.55}$. The interval does not contain 1, which agrees with the chi-square test.
- **Vaccine efficacy** = $(1 - RR) \times 100 = \mathbf{60\%}$ (95% CI 45% to 70%) against all reported polio. Against paralytic polio only (33 cases against 115), efficacy was 71%.
- **Risk difference** = $70.6 - 28.4 = 42.2$ fewer cases per 100,000 children vaccinated. About 2,400 children had to be vaccinated to prevent one case in that season.

The odds ratio here is also 0.40. When the outcome is rare, the odds ratio and the risk ratio are almost identical.

REAL EXAMPLE: READING A PUBLISHED RESULT

Real data

The Physicians' Health Study randomised 22,071 male doctors in the United States to 325 mg of aspirin every other day or placebo and followed them for about five years. The published result for a first heart attack was a relative risk of **0.56 (95% CI 0.45 to 0.70; $p < 0.00001$)**.

How to read it: aspirin users had 56% of the risk of heart attack, a 44% reduction. The whole interval lies well below 1, so the result is statistically clear, and the true reduction is plausibly anywhere from 30% to 55%. The same study found no reduction in deaths from cardiovascular disease overall, which is a reminder to look at all the important outcomes and not only the one with the striking p-value.

3.13 Checking assumptions and the multiple testing problem

Assumptions to check before trusting a test

Assumption	Needed by	How to check	If it fails
Independence of observations	Almost every test	Think about the design: clusters, households, repeated measurements?	Paired tests, mixed models, or survey methods
Normality of the data or of the sample means	t-tests, ANOVA, Pearson	Histogram, Q-Q plot, Shapiro-Wilk test	Rely on the CLT if n is large; otherwise transform (log) or use a rank test
Equal variances across groups	Student's t-test, classic ANOVA	Compare the SDs, Levene's test	Welch's t-test or Welch's ANOVA
Expected counts of at least 5	Chi-square test	Look at the expected table	Fisher's exact test, or merge categories
Straight-line relationship	Pearson correlation	Scatter plot	Spearman correlation

A warning about normality tests: with a small sample the Shapiro-Wilk test often fails to detect non-normal data, and with a very large sample it flags tiny departures that do not matter. Plots are usually more informative.

Multiple testing

Each test at $\alpha = 0.05$ has a 5% chance of a false alarm when there is no real effect. Run many tests and false alarms become almost certain.

$$P(\text{at least one false positive in } m \text{ independent tests}) = 1 - (1 - \alpha)^m$$

Number of tests	1	5	10	20
Chance of at least one false positive	5%	23%	40%	64%

A survey that tests 20 risk factors against one disease should expect about one "significant" association by chance alone. The simplest protection is the **Bonferroni correction**: use α/m as the cut-off, so $0.05 \div 20 = 0.0025$ for 20 tests. It is strict. Gentler methods exist (Holm, and the Benjamini-Hochberg false discovery rate). The better protection is to state one main question before the analysis and to label everything else as exploratory.

COMMON MISTAKE

p-hacking: trying many outcomes, subgroups, cut-offs or tests and reporting only the ones with $p < 0.05$. The reported p-values are then meaningless. Decide the analysis plan before looking at the results, and report every analysis you ran.

3.14 How to report and interpret results

A good results sentence contains four things: the estimate, its confidence interval, the test and p-value, and the meaning in plain words.

Weak	Better
"There was a significant difference in haemoglobin ($p < 0.05$)."	"Mean haemoglobin was 0.7 g/dL higher in the supplemented group (11.8 against 11.1 g/dL; 95% CI for the difference 0.05 to 1.35; Welch's t-test, $p = 0.034$)."
"Vaccination was associated with polio ($p = 0.000$)."	"Polio occurred in 28.4 per 100,000 vaccinated children and 70.6 per 100,000 in the placebo group (risk ratio 0.40, 95% CI 0.30 to 0.55; $p < 0.001$)."
"The intervention had no effect ($p = 0.21$)."	"The difference was 2.1 mmHg (95% CI -1.2 to 5.4; $p = 0.21$). The study could not rule out a reduction of up to 5 mmHg."

- Give exact p-values ($p = 0.034$), except write $p < 0.001$ for very small ones. Never write $p = 0.000$.
- Name the test you used and say whether it was two-sided.
- Report the number of people in each analysis.
- Say "not statistically significant", never "insignificant" or "no difference".
- Finish with the public health meaning: who is affected, by how much, and what should be done.

COMMON MISTAKES IN INFERENTIAL STATISTICS

- Using an independent t-test on paired data, or a paired test on independent groups.
- Running several t-tests instead of ANOVA when there are three or more groups.
- Using the chi-square test when expected counts are below 5.
- Treating "not significant" as proof of no effect, especially in a small study.
- Treating "significant" as proof of an important or causal effect.
- Reporting SE in place of SD to make the data look less variable.
- Ignoring clustering and weights when analysing survey data such as BDHS, which makes confidence intervals too narrow.
- Choosing the test after seeing which one gives the smallest p-value.
- Forgetting that statistics measures chance, not bias. A biased sample gives a precise wrong answer.

3.15 Check yourself: Part 3

1. A survey of 900 adults finds 180 with hypertension. Calculate the prevalence and its 95% confidence interval.

Prevalence = $180 \div 900 = 20\%$. $SE = \sqrt{(0.20 \times 0.80 \div 900)} = 0.0133$. Margin = $1.96 \times 0.0133 = 0.026$. 95% CI = 17.4% to 22.6%.

2. Which test would you use? (a) Mean birth weight in smokers' and non-smokers' babies. (b) Blood sugar in the same 30 patients before and after a diet. (c) Vaccination status (yes/no) by mother's education (4 levels). (d) Household income in urban and rural areas. (e) BMI across four wealth groups.

(a) Independent (Welch) t-test. (b) Paired t-test. (c) Chi-square test, or chi-square test for trend because education is ordered. (d) Mann-Whitney U test, because income is skewed. (e) One-way ANOVA, then a post hoc test.

3. A trial of 30 patients reports "no significant difference, $p = 0.18$ ". A colleague concludes the drug does not work. What is wrong?

A small study has low power, so a Type II error is likely. Look at the effect size and its confidence interval. If the interval includes clinically important benefits, the study is inconclusive, not negative.

4. A study reports an odds ratio of 1.8 with 95% CI 0.9 to 3.6. Is it statistically significant at the 5% level?

No. The interval contains 1, the "no effect" value for a ratio, so p is above 0.05. The data are compatible with anything from a small protective effect to a risk more than three times higher, so the study is imprecise.

5. You want to estimate exclusive breastfeeding prevalence, expected to be about 55%, within ± 4 percentage points at 95% confidence, using a cluster survey with design effect 2. What sample size do you need?

$n = 1.96^2 \times 0.55 \times 0.45 \div 0.04^2 = 3.8416 \times 0.2475 \div 0.0016 = 594.3$, rounded up to 595. With design effect 2: 1,190.

6. A researcher compares 12 biomarkers between cases and controls and finds one with $p = 0.03$. What should she conclude?

With 12 tests the chance of at least one false positive is $1 - 0.95^{12} = 46\%$. The Bonferroni cut-off is $0.05 \div 12 = 0.004$, and 0.03 does not pass it. The finding is a lead for a new study, not a conclusion.

Appendix: Formula Sheet, Tables, Glossary and Sources

A quick reference for revision. Every formula here is explained with an example in the section shown.

A.1 Formula sheet

Quantity	Formula	Section
Mean	$\bar{x} = \sum x_i / n$	1.5
Weighted mean	$\bar{x}_w = \sum w_i x_i / \sum w_i$	1.5
Variance and SD	$s^2 = \sum (x_i - \bar{x})^2 / (n - 1), s = \sqrt{s^2}$	1.6
Interquartile range	$IQR = Q_3 - Q_1$	1.6
Coefficient of variation	$CV = (s / \bar{x}) \times 100\%$	1.6
Outlier fences	$Q_1 - 1.5 IQR$ and $Q_3 + 1.5 IQR$	1.7
z-score	$z = (x - \mu) / \sigma$	1.9
Pearson correlation	$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$	1.10
Complement	$P(\text{not } A) = 1 - P(A)$	2.2
Addition rule	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$	2.2
Conditional probability	$P(A B) = P(A \text{ and } B) / P(B)$	2.3
At least one in n	$1 - (1 - p)^n$	2.3
Odds	$\text{odds} = p / (1 - p), p = \text{odds} / (1 + \text{odds})$	2.4
Bayes' theorem	$P(A B) = P(B A)P(A) / P(B)$	2.5
Sensitivity, specificity	$TP / (TP + FN), TN / (TN + FP)$	2.5
PPV, NPV	$TP / (TP + FP), TN / (TN + FN)$	2.5
Expected value, variance	$E(X) = \sum xP(x), \text{Var}(X) = E(X^2) - [E(X)]^2$	2.6
Binomial	$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}; \text{mean } np, \text{variance } np(1 - p)$	2.7
Poisson	$P(X = k) = e^{-\lambda} \lambda^k / k!; \text{mean } \lambda, \text{variance } \lambda$	2.8
Negative binomial	$\text{mean } \mu, \text{variance } \mu + \mu^2 / k$	2.9
Exponential	$P(T > t) = e^{-\lambda t}; \text{mean } 1 / \lambda$	2.11

Standard error of a mean	$SE = s/\sqrt{n}$	3.2
Standard error of a proportion	$SE = \sqrt{\hat{p}(1 - \hat{p})/n}$	3.2
Confidence interval	estimate $\pm$ critical value $\times$ SE	3.4
Sample size, one proportion	$n = z^2 p(1 - p)/d^2$	3.8
Sample size, two proportions	$n = (z_{\alpha/2} + z_{\beta})^2 [p_1(1 - p_1) + p_2(1 - p_2)] / (p_1 - p_2)^2$	3.8
Two-sample t (Welch)	$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{s_1^2/n_1 + s_2^2/n_2}$	3.10
Paired t	$t = \bar{d} / (s_d / \sqrt{n})$	3.10
Chi-square	$\chi^2 = \sum (O - E)^2 / E$, $E = \text{row} \times \text{column} / \text{total}$	3.11
Risk ratio, odds ratio	$RR = (a/n_1) / (c/n_0)$, $OR = ad/bc$	3.12
Bonferroni cut-off	α/m	3.13

A.2 Reference tables

Standard normal distribution: area to the left of z

z	-3.0	-2.5	-2.0	-1.96	-1.645	-1.5	-1.28	-1.0	-0.5	0
Area	0.0013	0.0062	0.0228	0.0250	0.0500	0.0668	0.1003	0.1587	0.3085	0.5000
z	0.5	1.0	1.28	1.5	1.645	1.96	2.0	2.5	2.576	3.0
Area	0.6915	0.8413	0.8997	0.9332	0.9500	0.9750	0.9772	0.9938	0.9950	0.9987

Area to the right of $z = 1 - \text{area to the left}$. Area between $-z$ and $+z = 2 \times (\text{area to the left of } z) - 1$.

t distribution: critical values for a 95% confidence interval (two-sided $\alpha = 0.05$)

Degrees of freedom	5	10	15	20	24	30	60	120	Very large
Critical t	2.571	2.228	2.131	2.086	2.064	2.042	2.000	1.980	1.960

A.3 Glossary

Term	Meaning in simple words
Alpha (α)	The accepted chance of a false alarm (Type I error), usually 0.05.
Bias	A systematic error that pushes results in one direction. Not reduced by a bigger sample.
Case fatality	The share of people with a disease who die from it.

Central limit theorem	Sample means are approximately normal when the sample is large, whatever the shape of the data.
Confidence interval	A range that is likely to contain the true population value.
Confounder	A third variable related to both the exposure and the outcome that distorts their relationship.
Degrees of freedom	The number of values that are free to vary once certain totals are fixed. It selects the right t, chi-square or F curve.
Design effect	How much a cluster survey inflates the variance compared with a simple random sample of the same size.
Effect size	How big a difference or association is.
Incidence	New cases of a disease in a population over a period of time.
Independence	Knowing one event tells you nothing about the other.
Null hypothesis	The statement of no effect or no difference that a test tries to reject.
Odds	The chance that something happens divided by the chance that it does not.
Outlier	A value far from the rest of the data.
Overdispersion	Count data whose variance is larger than the mean.
p-value	The probability of a result at least as extreme as the one observed, if the null hypothesis were true.
Parameter	A true but usually unknown number describing a population.
Percentile	The value below which a given percentage of the data fall.
Power	The chance that a study detects an effect that really exists.
Prevalence	The share of a population that has a condition at a given time.
Robust	Not strongly affected by extreme values.
Sampling error	The difference between a sample result and the truth that is due to chance alone.
Sensitivity	The share of diseased people that a test correctly identifies.
Skewness	Lack of symmetry: one tail of the distribution is longer.
Specificity	The share of healthy people that a test correctly identifies.
Standard deviation	The typical distance of individual values from the mean.
Standard error	The typical distance of a sample estimate from the true population value.
Statistic	A number calculated from a sample.
Type I error	Concluding there is an effect when there is none.
Type II error	Missing an effect that really exists.

z-score

How many standard deviations a value is from the mean.

A.4 Sources for the real data

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6. **Salk polio vaccine field trial, 1954**. Francis T. et al. An evaluation of the 1954 poliomyelitis vaccine trials. American Journal of Public Health, 1955; 45 (5, Part 2). Counts for the placebo-controlled areas as tabulated at randomservices.org/random/data/Polio.html
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8. **p-values**. Wasserstein R.L., Lazar N.A. The ASA Statement on p-Values: Context, Process, and Purpose. The American Statistician, 2016; 70: 129-133.
9. **Population of Bangladesh**. Bangladesh Bureau of Statistics, Population and Housing Census 2022 (about 170 million).

A.5 What comes next in the roadmap

Idea from these notes	Where it is used next
Proportions, rates, the 2×2 table, risk ratio	Epidemiology: prevalence, incidence, measures of association
Conditional probability, confounding	Bias, confounding and causality
Correlation, t-test, normal distribution	Linear regression
Binomial distribution, odds	Logistic regression and odds ratios
Poisson distribution, rate $\times$ exposure	Poisson regression and offsets
Overdispersion	Negative binomial regression
Exponential distribution, hazard	Survival analysis: Kaplan-Meier and Cox regression
Independence, design effect	Mixed-effects models and complex survey analysis